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Abstract

We propose a novel no-arbitrage framework, which exploits convex asset pricing constraints to study investors' marginal utility of wealth or, more generally, Stochastic Discount Factors (SDFs). We establish a duality between minimum dispersion SDFs and penalized portfolio selection problems, building the foundation for characterizing the feasible tradeoffs between a SDF's pricing accuracy and its comovement with systematic risks. Empirically, a minimum variance CAPM-SDF produces a Pareto optimal tradeoff. This SDF only depends on two distinct risk factors: A traded market factor and a minimum variance excess return that bounds the mispricing of risks unspanned by market shocks.

Keywords: SDF, Convex Pricing Constraints, Minimum Dispersion SDF, Market Frictions, SDF regularization, Arbitrage Pricing Theory,

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The absence of arbitrage opportunities is equivalent to the existence of a Stochastic Discount Factor (SDF), which can be interpreted as a proxy for the marginal utility of wealth in asset markets. Since SDFs are key for understanding asset prices, they have been the object of a vast amount of research, which builds on various ex-ante hypotheses regarding the SDF pricing features or, equivalently, the underlying market structure. This paper focuses on the SDF properties that can be learned from asset prices, based exclusively on the information conveyed by a set of convex asset pricing constraints in an arbitrage-free market.

We first establish the arbitrage-free foundation of SDFs defined by general convex pricing constraints and provide a novel unifying framework for analysing and selecting these SDFs. Importantly, some assets in our approach may only satisfy a set of convex pricing constraints with respect to a given SDF, e.g., because they may involve transaction costs in partially liquid markets or because their payoffs may reflect ambiguous risk exposures with uncertain compensations. Other assets may instead be (or be not) priced exactly by the same SDF, e.g., assets involving negligible transaction costs and corresponding to unambiguous systematic risk exposures. We call such SDFs Smart SDFs (S-SDFs), because of their affinity to the motivation of smart beta portfolios in modern asset management.¹

In arbitrage-free economies with market frictions, S-SDFs represent a positive linear pricing rule that bounds from below the nonlinear pricing rule of such markets. In frictionless arbitrage-free economies with ambiguity, S-SDFs represent the positive linear pricing rule of an unobserved marginal investor. This foundation provides a novel framework for several important asset pricing theories, such as Ross [1976]' Arbitrage Pricing Theory (APT), the good-deal bound SDFs in Cochrane and Saa-Requejo [2000], various approaches to SDF regularization such as Kozak et al. [2020], and robust approaches for the identification of investors' beliefs such as Chen et al. [2020].

¹Assets that are often assumed exactly priced for empirical asset pricing purposes are tradeable portfolios replicating established systematic risk exposures in the literature, such as market risk or other exposures like size and value, among others.

Our analysis and selection framework for S-SDFs is based on minimum dispersion S-SDF problems subject to convex pricing constraints. We establish a duality between minimum dispersion S-SDF problems and penalized dual portfolio selection problems, in which the penalization choice stays in a one-to-one relation with the initial pricing constraints. Since minimum dispersion S-SDFs are given by simple transformations of dual optimal portfolio payoffs, we find that they are naturally interpretable as regularized SDFs in an arbitrage-free economy with frictions or ambiguous asset payoffs.

Our framework gives rise to a novel methodology for characterizing key S-SDF properties, based on the attainable tradeoffs between pricing accuracy and S-SDF comovement with systematic asset return risks. This framework relies on a broad family of minimum variance S-SDFs that satisfy convenient asset pricing bounds consistent with the APT. Using cross-sections of sorted portfolio returns of varying dimensions, we take our theoretical APT S-SDF framework to market data. We find that a minimum variance correction of a CAPM-SDF, in which market risk alone is priced exactly and the mispricing of risks orthogonal to market risk is bounded according to APT pricing constraints, consistently produces a Pareto optimal tradeoff. In contrast, S-SDFs forcing tradability of additional systematic risk exposures uniformly imply a suboptimal tradeoff.

Our minimum variance Pareto optimal S-SDF corrections are founded in arbitrage-free frictionless markets with ambiguity and feature a particularly simple structure, which only depends on two economically interpretable asset pricing factors. The first factor is naturally a traded excess return that strongly correlates with pervasive asset return risks. The second factor is a minimum variance portfolio excess return, which bounds the mispricing of excess returns unspanned by the traded factor. This evidence demonstrates that a convenient family of minimum variance S-SDFs, which depend on just two economically interpretable asset pricing factors and are founded in corresponding arbitrage-free markets, attains a robust Pareto optimal tradeoff between pricing accuracy and S-SDF comovement with systematic asset return risks. This optimal tradeoff provides a natural

benchmark for assessing also empirical S–SDF specifications not included in our family of minimum variance S–SDFs.

The rest of the paper proceeds as follows. After a review of the related literature, in Section 2, we define S–SDFs and prove their existence in arbitrage-free economies. We then introduce minimum dispersion S–SDF problems, derive their dual penalized portfolio problems and show how optimal S–SDFs arise as simple transformations of the optimal penalized portfolio payoff. In Section 3, we revisit various asset pricing theories and settings that can be founded and understood within our S–SDF framework, while Section 4 explicitly applies our theory to specify a family of model-free asset pricing relations consistent with the APT. The empirical evidence on APT S–SDFs regarding the tradeoffs between cross-sectional pricing accuracy and S–SDF comovement with systematic risks, together with the logic behind different S–SDFs specification choices, is collected in Section 5. Section 6 concludes, while an Online Appendix collects various auxiliary results, additional empirical evidence, and the proofs of the formal statements in the main text.

1 Related Literature

Our paper is related to various important strands of the literature. A first strand studies the no-arbitrage foundation of strictly positive SDFs with various versions of the fundamental theorem of asset pricing. Ross [1978], Harrison and Kreps [1979], and Jouini and Kallal [1995] prove early versions of the fundamental theorem for frictionless economies and for economies with sublinear transaction costs, respectively. In our arbitrage-free foundation of S–SDFs, we borrow from the market structure in Jouini and Kallal [1995] and build a class of suitable economies, in which a strictly positive S–SDF exists if and only if markets are arbitrage-free. This result provides a general foundation for S–SDFs consistent with convex pricing constraints, with distinct economic interpretations in corresponding arbitrage-free economies with frictions or ambiguity.

A second prominent direction builds on the no-arbitrage foundation in Harrison and

Kreps [1979] and studies minimum dispersion SDFs and SDF bounds under exact pricing constraints. First introduced by Hansen and Jagannathan [1991] as key objects to compute minimum variance bounds giving rise to model-free diagnostics for asset pricing models, they have been exploited in Snow [1991], Stutzer [1995], Bansal and Lehmann [1997], Alvarez and Jermann [2005], Backus et al. [2014], and Almeida and Garcia [2016], among others, to develop further model diagnostics relying on notions of SDF dispersion different from variance. Luttmer [1996] is the first to consider minimum variance SDFs with conic pricing constraints originating from market frictions such as short-sale constraints or bid-ask spreads. Our framework substantially extends this literature to general S-SDF settings with convex pricing constraints, which allows us to embed in a single unifying framework many asset pricing approaches that are not covered by this literature.

Our S-SDF framework provides a distinct arbitrage-free foundation for various directions in the literature. We show that the good-deal bound SDFs in Cochrane and Saa-Requejo [2000] and the robust SDFs in Kozak et al. [2020] are minimum variance S-SDFs corresponding to a particular choice of convex pricing constraints, while the robust investor belief identified in Chen et al. [2020] is interpretable as a S-SDF minimizing a particular notion of stochastic dispersion relative to a benchmark asset pricing model. We further embed the predictions of Ross [1976] APT and its version with misspecification in Raponi et al. [2018] and Uppal et al. [2019] in our framework using a corresponding family of APT S-SDFs, which give rise to useful model-free interpretations of the APT predictions and new ways to implement them empirically.² We show that such APT S-SDFs optimally capture the tradeoff between S-SDF pricing accuracy and the S-SDF ability to span systematic asset return risks.

Our approach is linked also to recent directions applying penalization techniques from

²The formalizations of the APT in Hubermann [1982], Chamberlain [1983], Chamberlain and Rothschild [1983], Ingersoll [1984], and Uppal et al. [2019] are based on an asymptotic no-arbitrage condition that gives rise to bounded convex pricing constraints under a quadratic APT metric. This feature allows us to construct a family of minimum variance APT S-SDFs replicating the APT asset pricing predictions, under varying assumptions on the set of tradeable systematic risk factors and the S-SDF sparsity features.

the machine learning literature in empirical asset pricing. Freyberger et al. [2020] explain expected returns with a small set of characteristics based on a nonparametric specification of dependencies between expected returns and characteristics. Gu et al. [2020b] find evidence that machine learning methods allowing for nonlinear predictive relations outperform in out-of-sample return prediction. Feng et al. [2020] propose a robust model-selection method for evaluating the contribution of new factors to an unknown SDF. Different from this literature, we propose a unifying framework that directly constructs minimum dispersion S–SDFs with convex pricing properties from a cross-section of asset returns, thereby coherently addressing also settings where expected excess returns may not be fully explained by a covariation with a SDF.

A more recent literature studies empirical SDFs with improved out-of-sample pricing properties. Kozak et al. [2020] introduce versions of such SDFs, using lasso, ridge, and elastic net penalizations. We show that the population versions of these SDFs are interpretable as minimum dispersion S–SDFs and obtain in closed-form their hidden pricing features. Given such pricing features, we show that a Pareto optimal tradeoff between pricing accuracy and SDF comovement with systematic return risks is attained by minimum variance S–SDFs that depend on the excess returns of just two economically interpretable portfolios. Our framework provides a useful starting point also for empirical SDFs aiming to incorporate conditional information from a large set of conditioning variables, such as Gu et al. [2020a]. A common assumption in these approaches is existence of a SDF exactly pricing all assets. Distinctions between these approaches arise in the way how they regularize the very high-dimensional unconditional pricing constraints that need to hold theoretically. Here, our framework can offer guidelines for interpreting a penalization choice in terms of the resulting pricing constraints of a corresponding S–SDF, which may exist in markets of growing dimension when a SDF may not.

2 Theory of Smart Stochastic Discount Factors

Denote by $L^p(\mathbb{P})$ ($p > 1$) the payoff space of random variables defined on probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and having finite p -th moment, by $\mathcal{Z} \subset L^p(\mathbb{P})$ a market of traded payoffs, and by $\pi : \mathcal{Z} \rightarrow \mathbb{R}$ a pricing functional associating a price to each traded payoff.³ We denote the resulting economy, which may or may not be completely observed, by the triplet $(\mathcal{Z}, \pi, \mathbb{P})$. In such economy, we finally let $\mathbf{X} = (X_n)_{n=1}^N$ be a finite observed subset of N traded payoffs, with associated price vector $\mathbf{P} \in \mathbb{R}^N$.

2.1 S-SDF definition and basic properties

For any set of convex pricing constraints on payoff vector $\mathbf{X}$, we study the arbitrage-free market structures $(\mathcal{Z}, \pi, \mathbb{P})$ and corresponding stochastic discount factors compatible with such constraints. Without loss of generality, we specify convex pricing constraints on payoff vector $\mathbf{X}$ with a closed convex set $C \subset \mathbb{R}^N$ containing $\mathbf{0}$. This gives following definition of a smart stochastic discount factor.

Definition 1 (Smart Stochastic Discount Factors). Given constraint set C , a Smart Stochastic Discount Factor (S-SDF) is an element of following family of random variables:

$$\mathcal{M}_+(C) := \{M \geq 0 : \mathbb{E}[M\mathbf{X}] - \mathbf{P} \in C\} \cap L^q(\mathbb{P}), \quad (1)$$

where $q := p/(p-1)$.

In Definition 1, positive linear pricing functional $\mathbb{E}[M\cdot]$ gives rise to asset valuations constrained by set $\mathbf{P} + C := \{\mathbf{P} + \boldsymbol{\eta} : \boldsymbol{\eta} \in C\}$. Intuitively, set C fixes the geometry of the discrepancies between vector $\mathbf{P}$ and the vector of S-SDF asset valuations. It thus crucially determines key S-SDF properties, such as the structure of the resulting asset

³We consider power $p \notin \{1, \infty\}$ in order to obtain compact proofs based on $L^p - L^q$ duality, with $1/p + 1/q = 1$. In this setting, we also apply the standard convention that equalities, inequalities and strict inequalities are understood $\mathbb{P}$ -almost surely.

excess returns. In general, following S-SDF expected excess return identity holds for any asset payoff:

$$\mathbb{E}[X_n] - \frac{P_n}{\mathbb{E}[M]} = -Cov \left[\frac{M}{\mathbb{E}[M]}, X_n \right] + \frac{\mathbb{E}[MX_n] - P_n}{\mathbb{E}[M]} . \quad (2)$$

Therefore, the choice of constraint set C directly determines the way how a S-SDF explains asset excess returns under a given reference probability $\mathbb{P}$. For instance, while S-SDFs completely explain the excess return of exactly valued assets by their covariance with S-SDF risk, only a part of the expected excess return of assets that are not exactly valued is explained by an exposure to S-SDF risk.

2.2 Existence of S-SDFs

Given price vector $\mathbf{P}$ and constraint set C , a key question is when does a S-SDF exist. We answer this question by explicitly constructing an arbitrage-free economy $(\mathcal{Z}_C, \pi_C, \mathbb{P})$ in which the S-SDF existence is granted. In such economy, investors can form portfolios $\boldsymbol{\theta} \in \mathbb{R}^N$ of payoffs $\mathbf{X}$, while incurring costs for including assets in a portfolio. These costs are described by a cost function defined by:

$$\sigma_C(\boldsymbol{\theta}) := \sup\{\boldsymbol{\theta}'\boldsymbol{\eta} : \boldsymbol{\eta} \in C\} \geq 0 . \quad (3)$$

Traded payoffs are those that can be replicated with finite costs:

$$\mathcal{Z}_C := \{Z = \mathbf{X}'\boldsymbol{\theta} : \sigma_C(\boldsymbol{\theta}) < +\infty\} . \quad (4)$$

Finally, the price of a traded payoff is the lowest total replication cost across payoff replicating portfolios:

$$\pi_C(Z) := \inf_{\boldsymbol{\theta} \in \mathbb{R}^N} \{\mathbf{P}'\boldsymbol{\theta} + \sigma_C(\boldsymbol{\theta}) : Z = \mathbf{X}'\boldsymbol{\theta}\} . \quad (5)$$

Definitions (4)–(5) give rise to an explicit market structure $(\mathcal{Z}_C, \pi_C, \mathbb{P})$ embedding traded payoff vector $\mathbf{X}$. As the next remark underscores, such a market structure is general and

uniquely determined by constraint set C .

Remark 1. By definition, cost function (3) is closed and sublinear.⁴ Conversely, any closed sublinear cost function σ can be uniquely written in the form (3) for a corresponding closed convex set $C := C_\sigma$, given explicitly by (see [Hiriart-Urruty and Lemaréchal, 2012, Thm. 3.1.1]):

$$C_\sigma := \{\boldsymbol{\eta} \in \mathbb{R}^N : \boldsymbol{\eta}'\boldsymbol{\theta} \leq \sigma(\boldsymbol{\theta}) \text{ for all } \boldsymbol{\theta} \in \mathbb{R}^N\} . \quad (6)$$

The uniquely associated economy (4)–(5) to this cost function is $(\mathcal{Z}_{C_\sigma}, \pi_{C_\sigma}, \mathbb{P})$. In such economies, differences between $\pi_C(X_n)$ and P_n may arise when cost function σ_C does not vanish on the unit vectors in $\mathbb{R}^N$, and they are uniquely determined by the properties of constraint set C .

The next proposition characterizes the existence of a strictly positive S–SDF by the absence of arbitrage in market $(\mathcal{Z}_C, \pi_C, \mathbb{P})$, where arbitrage opportunities are defined as free lunches following Harrison and Kreps [1979].⁵

Proposition 1. *Market $(\mathcal{Z}_C, \pi_C, \mathbb{P})$ admits no free lunches if and only there exists $M \in \mathcal{M}_+(C)$ such that $M > 0$.*

From Proposition 1, a strictly positive S–SDF exists if and only if market $(\mathcal{Z}_C, \pi_C, \mathbb{P})$ is arbitrage-free. Together with Remark 1, this provides a general arbitrage-free foundation for S–SDFs.

Remark 2. Proposition 1 in Online Appendix B.4 proves a slightly more general version of Proposition 1 for a market structure (4)–(5) with closed convex (not necessarily sublinear) cost function σ . In this context, existence of a S–SDF is equivalent to the viability of market $(\mathcal{Z}_{C_\sigma}, \pi_{C_\sigma}, \mathbb{P})$, which is in general a stronger no arbitrage condition than the absence of free lunches.⁶

⁴A mapping $f : V \rightarrow \mathbb{R}$ is closed if sublevel set $\{x \in V : f(x) \leq \alpha\}$ is closed for any $\alpha \in \mathbb{R}$. It is sublinear if $f(\alpha x) = \alpha f(x)$ and $f(x + y) \leq f(x) + f(y)$ for any $\alpha \geq 0$ and $x, y \in V$.

⁵Formally, a free lunch in market $(\mathcal{Z}, \pi, \mathbb{P})$ consists of a sequence $\{Y_n\} \subset L^p(\mathbb{P})$ such that $Y_n \rightarrow Y \geq 0$ and a sequence $\{Z_n\} \subset \mathcal{Z}$ such that $\liminf_n \pi(Z_n) \leq 0$ and $Z_n \geq Y_n$ for every $n \in \mathbb{N}$.

⁶A market $(\mathcal{Z}, \pi, \mathbb{P})$ is viable if there exists an agent (represented by preference $\succsim$) and $Z^* \in \mathcal{Z}$ such

2.3 Foundation in markets with frictions or ambiguity

Our S–SDF framework embeds most existing asset pricing settings. An obvious one is the frictionless arbitrage-free framework of Hansen and Jagannathan [1991], in which $C = \{\mathbf{0}\}$, $\mathcal{Z} = \{\boldsymbol{\theta}'\mathbf{X} : \boldsymbol{\theta} \in \mathbb{R}^N\}$ and price vector $\mathbf{P}$ is matched exactly by the vector of asset valuations of a strictly positive SDF. Market settings with frictions corresponding to sublinear or convex transaction cost functions σ are naturally embedded as well. Here, Proposition 1 and Proposition 1 of Online Appendix B.4 provide the unique constraint set $C := C_\sigma$ of a strictly positive S–SDF associated to sublinear or convex specifications of transaction costs σ .⁷

It is important to note that S–SDFs can arise also in frictionless markets, when the unobserved marginal investor belief and the probability belief $\mathbb{P}$ used to describe asset payoffs are not equivalent, offering a convenient framework to formulate robust asset pricing predictions in a variety of settings where an SDF may not exist under belief $\mathbb{P}$.⁸ To clarify this, denote by $\tilde{\mathbb{P}}$ the marginal investor belief and let frictionless market $(\tilde{Z}_{\{\mathbf{0}\}}, \tilde{\pi}_{\{\mathbf{0}\}}, \tilde{\mathbb{P}})$ admit no free lunches. From Proposition 1, there exists a SDF $\tilde{M}$ such that:

$$\tilde{\mathbb{E}}[\tilde{M}\mathbf{X}] = \mathbf{P} , \quad (7)$$

where $\tilde{\mathbb{E}}[\cdot]$ denotes expectations under probability $\tilde{\mathbb{P}}$. Importantly, an exact price representation of the form (7) can hold for a strictly positive SDF M with respect to a probability belief $\mathbb{P}$ if and only $\mathbb{P}$ and $\tilde{\mathbb{P}}$ are equivalent, in which case:

$$\mathbf{P} = \tilde{\mathbb{E}}[\tilde{M}\mathbf{X}] = \mathbb{E}[M\mathbf{X}] , \quad (8)$$

that $\pi(Z^*) \leq 0$ and $Z^* \succsim Z$ for all $Z \in \mathcal{Z}$ with $\pi(Z) \leq 0$. The preference relation $\succsim$ is assumed to be convex, continuous and strictly increasing; see again Harrison and Kreps [1979].

⁷For instance, economies with frictions given by short-sale constraints or bid-ask spreads arise for $C := -\mathbb{R}_+^N$ (the negative cone in $\mathbb{R}^N$) and a set $\mathcal{Z} = \{\boldsymbol{\theta}'\mathbf{X} : \boldsymbol{\theta} \in \mathbb{R}_+^N\}$ of traded payoffs with cost function $\sigma(\boldsymbol{\theta}) = 0$ ($\sigma(\boldsymbol{\theta}) = +\infty$) when $\boldsymbol{\theta} \geq \mathbf{0}$ (else); see again Luttmer [1996].

⁸For instance, in all continuous-time theoretical asset pricing settings, non equivalent beliefs $\tilde{\mathbb{P}}$ and $\mathbb{P}$ arise whenever the associated return volatility processes are not identical.

with the Radon-Nykodim derivative $N := \frac{d\tilde{\mathbb{P}}}{d\mathbb{P}}$ and $M := \tilde{M}N$. Conversely, no strictly positive SDF satisfying equation (8) under probability belief $\mathbb{P}$ can exist when some zero probability assessments under $\mathbb{P}$ and $\tilde{\mathbb{P}}$ differ. However, from Proposition 1 we also know more generally that for any closed convex set C containing $\{\mathbf{0}\}$ market $(\mathcal{Z}_C, \pi_C, \mathbb{P})$ admits no free lunches if and only if a strictly positive S–SDF M exists such that:

$$\mathbb{E}[M\mathbf{X}] - \tilde{\mathbb{E}}[\tilde{M}\mathbf{X}] \in C . \quad (9)$$

Here, set C can be interpreted as an implicit constraint on the degree of ambiguity between beliefs $\tilde{\mathbb{P}}$ and $\mathbb{P}$. Under such constraint, SDF $\tilde{M}$ and S–SDF M may imply distinct asset valuations under the corresponding beliefs, however with differences restricted to belong to set C . Such differences arise from the non-equivalence of $\mathbb{P}$ and $\tilde{\mathbb{P}}$, which renders the set of arbitrage opportunities perceived under each of these beliefs different.

Corollary 1 (S–SDFs and ambiguity). *Markets $(\tilde{\mathcal{Z}}_{\{\mathbf{0}\}}, \tilde{\pi}_{\{\mathbf{0}\}}, \tilde{\mathbb{P}})$ and $(\mathcal{Z}_C, \pi_C, \mathbb{P})$ admit no free lunches if and only if there exist strictly positive SDF $\tilde{M}$ and strictly positive S–SDF M such that constraints (9) hold, with pricing functional π_C explicitly given for any $Z \in \mathcal{Z}_C$ by:*

$$\pi_C(Z) = \inf_{\boldsymbol{\theta} \in \mathbb{R}^N} \left\{ \tilde{\mathbb{E}}[\tilde{M}\mathbf{X}'\boldsymbol{\theta}] + \sup_{\boldsymbol{\eta} \in C} \{\boldsymbol{\eta}'\boldsymbol{\theta}\} : \mathbb{P}(Z = \mathbf{X}'\boldsymbol{\theta}) = 1 \right\} .$$

π_C in Corollary 1 is interpretable as the min-max replication cost of a payoff Z , after considering all ambiguous asset valuations under belief $\mathbb{P}$, which result from admissible price adjustments $\boldsymbol{\eta}$ relative to the marginal investor valuation $\mathbf{P} = \tilde{\mathbb{E}}[\tilde{M}\mathbf{X}]$. Since the no free lunches condition in market $(\mathcal{Z}_C, \pi_C, \mathbb{P})$ is weaker than the no free lunches condition in market $(\mathcal{Z}_{\{\mathbf{0}\}}, \pi_{\{\mathbf{0}\}}, \mathbb{P})$, because by definition $\mathcal{Z}_{\{\mathbf{0}\}} \supset \mathcal{Z}_C$ and $\pi_C(Z) \geq \pi_{\{\mathbf{0}\}}(Z)$ for all $Z \in \mathcal{Z}_C$, the pricing predictions of S–SDFs in ambiguous frictionless settings are also more robust than those of SDFs in unambiguous frictionless settings.

2.4 Minimum Dispersion S–SDFs

Having established their existence in arbitrage-free economies, we next provide an analysis and selection framework for S–SDFs, by minimizing established notions of stochastic dispersion in the literature. We measure dispersion via ϕ –dispersions defined by integral functionals of the form $M \mapsto \mathbb{E}[\phi(M)]$, for a function $\phi : \mathbb{R} \rightarrow (-\infty, +\infty]$ such that ϕ_+ , the restriction of ϕ to the nonnegative real line, is closed and strictly convex with $(0, +\infty) \subset \text{dom}(\phi_+)$.^{9,10}

Definition 2 (Minimum dispersion S–SDFs). A minimum dispersion S–SDF M_0 is a S–SDF minimizing a particular ϕ –dispersion, i.e., it solves the minimization problem:¹¹

$$\Pi(C) := \inf_{M \in \mathcal{M}_+(C)} \mathbb{E}[\phi(M)] . \quad (10)$$

By definition, minimum dispersion S–SDF problems in Definition 2 give rise to an extended family of SDF dispersion bounds, relative to the bounds in Hansen and Jagannathan [1991], Luttmer [1996] and Almeida and Garcia [2016], among others, which can account for general convex pricing constraints. Since working directly with problem (10) is inconvenient, as it is an infinite-dimensional optimization problem, we provide in the sequel the dual characterization of minimum dispersion S–SDFs and S–SDF dispersion bounds via the solution of following penalized portfolio problem:

$$\Delta(C) := \min_{\theta \in \mathbb{R}^N} \{ \mathbb{E}[\phi_+^*(-\mathbf{X}'\theta)] + \mathbf{P}'\theta + \sigma_C(\theta) \} = \min_{\theta \in \mathbb{R}^N} \{ \mathbb{E}[\phi_+^*(-\mathbf{X}'\theta)] + \pi_C(\mathbf{X}'\theta) \} , \quad (11)$$

with the penalization function σ_C in equation (3) and the convex conjugate of ϕ_+ , defined for any $y \in \mathbb{R}$ by $\phi_+^*(y) := \sup_{x \in \mathbb{R}} \{yx - \phi_+(x)\}$. In this portfolio problem, the dual

⁹ $\phi_+(x)$ is defined as $\phi_+(x) = \phi(x)$ when $x \geq 0$ and $\phi_+(x) = +\infty$ otherwise.

¹⁰Many well-known measures of SDF dispersion are ϕ –dispersions, such as the variance, entropy-based dispersions and more generally dispersions in the Cressie and Read [1984] family. Online Appendix B.1 collects relevant explicit examples of ϕ –dispersions in this family.

¹¹Naturally, the ϕ –dispersion in problem (10) is chosen so that $\phi(M) \in L_1$ for any $M \in \mathcal{M}_+(C) \subset L_q$, where $q = p/(p-1)$ and L_p ($p \in (1, \infty)$) is the underlying payoff space.

dispersion functional $Z \mapsto \mathbb{E}[\phi_+^*(Z)]$ measures the dispersion of portfolio payoffs Z , while $\sigma_C(\theta)$ penalizes a candidate assets' portfolio weight relative to the benchmark portfolio cost $\mathbf{P}'\theta$. Equivalently, in the second identity of equation (11), a candidate portfolio payoff is penalized by its total cost $\pi_C(\mathbf{X}'\theta)$ in the economy $(\mathcal{Z}_C, \pi_C, \mathbb{P})$ of Section 2.2.

The strong duality between minimum dispersion S-SDFs and optimal portfolios solving penalized problem (11) is established next.

Proposition 2 (Duality). *For some convex closed set $\tilde{C} \subset \mathbb{R}^N$ containing $\mathbf{0}$ consider the corresponding economy $(\mathcal{Z}_{\tilde{C}}, \pi_{\tilde{C}}, \mathbb{P})$ from Section 2.2. If such economy admits no free lunches, then $\Pi(C) = -\Delta(C)$ for any closed convex set C with a relative interior containing $\tilde{C}$.¹² If additionally θ_0 is a solution of problem (11) such that $-\mathbf{X}'\theta_0 < \lim_{y \rightarrow \infty} \phi(y)/y$, $\mathbb{P}$ -almost surely, then either minimum S-SDF dispersion problem (10) has no solution, or it has a unique solution given by:¹³*

$$M_0 = (\phi_+^*)'(-\mathbf{X}'\theta_0) . \quad (12)$$

Proposition 2 states that under appropriate conditions penalized portfolio problem (11) can be used to compute minimum dispersion bound (10) and minimum dispersion S-SDF M_0 , as a simple transformation of the optimal dual portfolio payoff in equation (12).¹⁴ Therefore, it provides a powerful device to compute minimum dispersion S-SDFs and S-SDF bounds in a wide variety of practically relevant asset pricing settings.

Remark 3. The duality result in Proposition 2 holds under a no-arbitrage condition of the form given in Proposition 1, for a set $\tilde{C}$ contained in the relative interior of constraint set C . This condition covers all practically relevant constraint sets C considered in this paper. For instance, if C is polyhedral as in, e.g., Luttmer [1996], then we can set $\tilde{C} = C$ without loss

¹²The relative interior of a set is the interior within the set's affine hull.

¹³With a slight abuse of notation, we denote by $(\phi_+^*)'(y)$ in equation (12) the first derivative of function ϕ_+^* in an interior point y of the domain of ϕ_+^* .

¹⁴Online Appendix B.1 collects closed-form expressions of ϕ_+ , ϕ_+^* and the minimum dispersion S-SDF (12) for the family of Cressie and Read [1984] dispersions.

of generality. Analogously, if C is a closed norm ball then $\tilde{C}$ can be any closed convex set included in the interior of C . Furthermore, if C is polyhedral or σ_C real-valued, then the no free lunches condition in Proposition 2 is equivalent to the standard no-arbitrage condition in the literature.¹⁵ Finally, the technical condition $-\mathbf{X}'\boldsymbol{\theta}_0 < \lim_{y \rightarrow \infty} \phi(y)/y$, $\mathbb{P}$ -almost surely, is obviously satisfied by any ϕ -dispersions function with superlinear growth, such as all Cressie-Read dispersions with power parameter $\gamma \geq 1$ in Online Appendix B.1.

Remark 4. When duality does not hold for a given closed convex set C ($\Pi(C) \neq \Delta(C)$), then for all closed convex sets $\tilde{C}$ included in the relative interior of C market $(\mathcal{Z}_{\tilde{C}}, \pi_{\tilde{C}}, \mathbb{P})$ is not arbitrage-free. In such situations, a solution to dual portfolio problem $\Delta(C)$ in equation (11) may still exist, but random variable M_0 in equation (12) may not be a minimum dispersion S-SDF. Emergence of such a duality failure can be assessed empirically, by verifying whether M_0 satisfies the pricing constraints in the primal minimum S-SDF dispersion problem (10). We borrow from this insight in our empirical analysis of Section 5, in order to distinguish asset pricing settings with solid S-SDF foundation from those with weak S-SDF foundation.

3 S-SDF Applications

Minimum dispersion S-SDFs and their dual portfolio problems are directly related to various important asset pricing settings corresponding to concrete equivalent specifications of constraint set C or penalty function σ_C in identity (3); see Table 1 for a list of closed-form (C, σ_C) pairs. These include, e.g., recent settings proposing SDFs that are regularized using techniques from the machine learning literature.

¹⁵See Theorem 6 in Clark [1993]. Formally, an arbitrage opportunity in a market $(\mathcal{Z}, \pi, \mathbb{P})$ is a nonzero traded payoff $Z \in \mathcal{Z}$ such that $X \geq 0$ and $\pi(Z) \leq 0$.

3.1 S–SDFs with bounded valuations

Minimum variance S–SDFs incorporating conic portfolio constraints, such as short sale constraints or bid-ask spreads, are studied in [Luttmer \[1996\]](#). As detailed in Online Appendix [B.2](#), such S–SDFs are obtained with Proposition [2](#) under a penalization implying zero (infinite) transaction costs for each portfolio satisfying (violating) the constraints. A specific property of these S–SDFs is that S–SDF asset valuations, while delimited by a convex cone, are not bounded. In the sequel, we address instead relevant asset pricing settings with bounded asset valuations.

A useful class of costs for the portfolio positions on dubious assets can be modeled using norms: $\sigma = \tau \|\cdot\|$, where $\|\cdot\|$ is some norm in $\mathbb{R}^N$ and $\tau > 0$ is a scaling parameter. With Proposition [1](#), S–SDFs in these markets give rise to bounded asset valuations based on the dual norm $\|\cdot\|_*$, since:^{[16](#)}

$$C_\sigma = \{\eta \in \mathbb{R}^N : \eta' \theta \leq \tau \|\theta\| \text{ for all } \theta \in \mathbb{R}^N\} = \{\eta \in \mathbb{R}^N : \|\eta\|_* \leq \tau\} . \quad (13)$$

For instance, a specification $\sigma = \tau \|\cdot\|_1$ of proportional transaction costs with an l_1 –norm implies differences between S–SDF valuations and vector $\mathbf{P}$ of prices that are uniformly bounded in absolute value by τ .^{[17](#)}

For asset pricing settings with no explicit role for transaction costs, S–SDFs implying norm bounded asset valuations can be founded in frictionless arbitrage-free economies with ambiguity. In such settings, a constraint set of the form [\(13\)](#) simply restricts the discrepancies between the marginal investors' arbitrage-free valuations of asset payoffs and the S–SDF valuations under belief $\mathbb{P}$.

Remark 5. Note that specification $\sigma_C = \tau \|\cdot\|_1$ with an l_1 –norm reproduces the widely

¹⁶The dual norm $\|\cdot\|_*$ is defined by $\|\eta\|_* := \max_{\theta \in \mathbb{R}^N} \{\eta' \theta : \|\theta\| \leq 1\}$.

¹⁷Given an l_p –norm such that $\|\mathbf{x}\|_p := \left(\sum_{i=1}^{N_D} |x_i|^p\right)^{1/p}$ when $p \in [1, +\infty)$ and $\|\mathbf{x}\|_p := \max_{i=1}^{N_D} |x_i|$ when $p = +\infty$, its dual norm is the l_q –norm with $1/p + 1/q = 1$. In particular, the l_∞ –norm is the dual norm of the l_1 –norm and vice-versa, while the l_2 –norm is self-dual.

used lasso penalty in the machine learning literature; see Tibshirani [1996]. This penalty is known to produce sparse optimal solutions, which in our setting implies sparse optimal portfolio weights in the solution of the dual portfolio problem of Proposition 2. Given the link (12) between optimal portfolios and minimum dispersion S–SDFs, this choice translates to S–SDFs that depend on a strict subset of asset returns. According to Proposition 2, these penalization techniques give rise to robust S–SDFs with pricing constraints defined by closed-form bounds of the form (13). Therefore, they also admit an explicit no-arbitrage foundation within a frictionless economy with ambiguity.

Remark 6. An important implication of our theory is that sparsity cannot be obtained together for the optimal portfolio weights of minimum dispersion S–SDFs and the vector $\boldsymbol{\eta}_0 = \mathbb{E}[M_0 \mathbf{X}] - \mathbf{P}$ of asset misvaluations under such SDF. This sparsity trade-off is general and arises also for penalizations implying a more flexible relation between shrinkage and sparsity than the lasso penalty. One such example is the Elastic Net penalty used in Kozak et al. [2020] to shrink the cross-section of returns. In our framework, a flexible relation between shrinkage and sparsity is obtained with norm-based penalizations combining the sparsity properties of the lasso and the shrinkage properties of l_2 –penalties: $\sigma_C = \alpha \|\cdot\|_1 + \tau \|\cdot\|_2$. The associated closed-form constraint set C in Table 1 bounds with a threshold τ the Euclidean distance of vector $\boldsymbol{\eta}_0$ from the l_∞ –norm ball of radius α . Here, maximal differences in S–SDF asset valuations larger than α are penalized quadratically, which gives rise to a quadratic constraint above a threshold that implies no sparsity in vector $\boldsymbol{\eta}_0$.¹⁸

3.2 Minimum dispersion S–SDFs as pricing error minimizers

As formally shown in Proposition OA-1 of Online Appendix A, minimum dispersion S–SDFs are equivalently defined by means of a minimum pricing error problem with bounded

¹⁸The limit case $\alpha = 0$ corresponds to a standard ridge pricing metric with penalization $\sigma_C = \tau \|\cdot\|_2$, which gives rise to non sparse optimal dual portfolio weights and a non sparse vector $\boldsymbol{\eta}_0$. Many other pricing metrics and portfolio weight penalties can be implemented in our framework, covering all sublinear penalties corresponding to any convex and closed constraint set C . Online Appendix B.4 shows how convex, but not sublinear, penalizations are covered as well by our theory.

S–SDF dispersion, using a convex and closed pricing error metric $h := h_C$ and a threshold $\tau := \tau_C$ such that $C = \{\boldsymbol{\eta} \in \mathbb{R}^N : h(\boldsymbol{\eta}) \leq \tau\}$. Such equivalent minimum pricing error problem reads:

$$\tilde{\Pi}(C) := \inf_{M \in L_+^q} \{h(\mathbb{E}[M\mathbf{X} - \mathbf{P}]) : \mathbb{E}[\phi(M)] \leq \nu\} , \quad (14)$$

for a corresponding bound $\nu := \nu_C > 0$ on the S–SDF dispersion. This equivalence establishes a useful link between our framework and further directions in the literature, such as the good-deal bounds theory in [Cochrane and Saa-Requejo \[2000\]](#), the robust investor’s belief identification approach in [Chen et al. \[2020\]](#) and the regularized SDFs proposed by [Kozak et al. \[2020\]](#).¹⁹ Given a vector $\mathbf{R}^e$ of N_D excess returns, benchmark examples of the latter are parametric S–SDFs of the form $M(\boldsymbol{\theta}_0) := \theta_{0S} - \boldsymbol{\theta}'_{0D}(\mathbf{R}^e - \mathbb{E}[\mathbf{R}^e])$, with $\boldsymbol{\theta}_0 = (\theta_{0S}, \boldsymbol{\theta}'_{0D})'$, and such that:

$$\boldsymbol{\theta}_0 = \arg \min_{\boldsymbol{\theta} \in \mathbb{R}^{N_D+1}} \{ \|\mathbb{E}[M(\boldsymbol{\theta})\mathbf{R}^e]\|_2 : \mathbb{E}[M^2(\boldsymbol{\theta})] \leq \nu \text{ and } \mathbb{E}[M(\boldsymbol{\theta})] = 1 \} .$$

This problem is naturally embedded into our framework, using asset payoff vector $\mathbf{X} := (1, \mathbf{R}^e)'$, price vector $\mathbf{P} = (1, \mathbf{0})'$ and constraint set $C = \{0\} \times \{\boldsymbol{\eta} \in \mathbb{R}^{N_D} : \|\boldsymbol{\eta}\|_2 \leq \tau\}$. Therefore, $M(\boldsymbol{\theta}_0)$ is a minimum variance S–SDF, with a norm-based pricing constraint of the form introduced in Section 3.1 and a dispersion function given explicitly by $\phi(x) = x^2$ for any $x \in \mathbb{R}$.

4 APT S–SDFs

The key constraint underlying the APT is a finite maximal Sharpe ratio in markets with no asymptotic arbitrage opportunities.²⁰ When returns satisfy a suitable factor structure, such a constraint gives rise to approximate linear valuation rules with asset pricing predictions that are reproducible by suitable S–SDFs with a norm bounded constraint set C . This

¹⁹See Online Appendix B.3 for a detailed derivation of the relation between our S–SDF methodology, good deal bounds theories and robust belief identification approaches.

²⁰See, e.g., [\[Chamberlain and Rothschild, 1983, Corollary 1\]](#) for a formal statement.

section explains the properties of these S-SDFs.

4.1 APT setting

Consider a factor model for a $N_D \times 1$ vector of excess returns $\mathbf{R}_{N_D}^e$:

$$\mathbf{R}_{N_D}^e = \mathbf{\Lambda}_{N_D} \mathbf{F}^e + \boldsymbol{\zeta}_{N_D} , \quad (15)$$

where $\mathbf{F}^e$ is a $N_S \times 1$ vector of observed traded excess factor returns and $\mathbf{\Lambda}_{N_D}$ a $N_D \times N_S$ matrix of factor loadings. The $N_D \times 1$ vector of residuals $\boldsymbol{\zeta}_{N_D}$ is orthogonal to traded factor risk, but potentially cross-sectionally correlated, with variance covariance matrix $\boldsymbol{\Sigma}_{\boldsymbol{\zeta}_{N_D}} = \mathbf{B}_{N_D} \mathbf{B}_{N_D}' + \mathbf{Q}_{N_D}$, where $\{\mathbf{B}_{N_D}\}$ is a sequence of $N_D \times N_K$ matrices and $\{\mathbf{Q}_{N_D}\}$ a sequence of symmetric positive definite $N_D \times N_D$ matrices with uniformly bounded eigenvalues.²¹ In this setting, it follows from [Chamberlain and Rothschild, 1983, Corollary 2] that the absence of asymptotic arbitrage opportunities, under a sequence of factor models (15) with growing dimension N_D , yields existence of a constant $\tau \geq 0$ and a $N_K \times 1$ vector $\boldsymbol{\lambda}$ such that:²²

$$\|\boldsymbol{\eta}_{N_D}\|_{2, \boldsymbol{\Sigma}_{\boldsymbol{\zeta}_{N_D}}^{-1/2}} := \sqrt{\boldsymbol{\eta}_{N_D}' \boldsymbol{\Sigma}_{\boldsymbol{\zeta}_{N_D}}^{-1} \boldsymbol{\eta}_{N_D}} \leq \tau , \quad (16)$$

where

$$\boldsymbol{\eta}_{N_D} := \mathbb{E}[\boldsymbol{\zeta}_{N_D}] - \mathbf{B}_{N_D} \boldsymbol{\lambda} . \quad (17)$$

In equation (17), $\boldsymbol{\eta}_{N_D}$ is an expected excess return component not explained by priced exposures to traded and unobservable factor risk in model (15).²³ Therefore, as stressed by Uppal et al. [2019], these predictions hold also in the case of a misspecified factor model

²¹Sufficient conditions for existence and uniqueness of such a factor representation are provided in [Chamberlain and Rothschild, 1983, Theorem 4].

²²See also Uppal et al. [2019]. Constant τ in inequality (16) is explicitly given as $\delta^2 \lambda_{k+1} / \lambda_0$, where $\lambda_{k+1} < \infty$ ($\lambda_0 > 0$) is a uniform upper (lower) bound on the $k+1$ largest (the smallest) eigenvalue of $\boldsymbol{\Sigma}_{\boldsymbol{\zeta}_N}$ and δ^2 is the squared maximal Sharpe ratio in the underlying arbitrage-free economy.

²³Since bound (16) is independent of N_D , we consider in the sequel an excess return vector $\mathbf{R}^e$ of fixed dimension and drop the N_D subscripts.

(15). On the other hand, $\mathbb{E}[\mathbf{F}^e]$ is the vector of risk premia of traded factors, which by construction have a price of zero.

4.2 Existence of APT–SDFs

The next corollary to Proposition 1 finds a family of S–SDFs compatible with APT pricing constraints (16), based on a founding economy that admits no arbitrage opportunities in the standard sense.²⁴

Corollary 2 (APT S–SDF). *Given the above APT setting, consider constraint set $C = \{\mathbf{0}_{N_S+1}\} \times \{\boldsymbol{\eta} \in \mathbb{R}^{N_D} : \|\boldsymbol{\eta}\|_{2, \boldsymbol{\Sigma}_\zeta^{-1/2}} \leq \tau\}$ and the associated penalization σ_C , which for any $\boldsymbol{\theta} = (\theta, \boldsymbol{\theta}'_S, \boldsymbol{\theta}'_D)'$ is given in closed-form by:*

$$\sigma_C(\boldsymbol{\theta}) = \tau \|\boldsymbol{\theta}_D\|_{2, \boldsymbol{\Sigma}_\zeta^{1/2}} . \quad (18)$$

Then, market $(\mathcal{Z}_C, \pi_C, \mathbb{P})$ in equations (4)–(5) is arbitrage-free if and only if there exists a strictly positive S–SDF M_{APT} , called APT S–SDF, with normalized expectation $\mathbb{E}[M_{APT}] = 1$ and such that:

$$\mathbb{E}[M_{APT} \mathbf{F}^e] = \mathbf{0} \text{ and } \|\mathbb{E}[M_{APT} \mathbf{R}^e]\|_{2, \boldsymbol{\Sigma}_\zeta^{-1/2}} \leq \tau . \quad (19)$$

Corollary 2 provides a model-free foundation of APT asset pricing relations in a frictionless arbitrage-free economy with ambiguity, where the valuation of excess returns orthogonal to traded factor risks is constrained with the standardized APT metric $h = \|\cdot\|_{2, \boldsymbol{\Sigma}_\zeta^{-1/2}}$. Indeed, under the APT S–SDF in Corollary 2 traded factor risks obey the standard identity:

$$\mathbb{E}[\mathbf{F}^e] = -Cov(M_{APT}, \mathbf{F}^e) , \quad (20)$$

i.e., tradable factor risk premia are completely explained by the exposure of traded factor

²⁴See again Footnote 15 for the formal definition of an arbitrage opportunity.

excess returns to APT S-SDF risk. In contrast, excess returns not fully spanned by traded factor risk satisfy the identity:

$$\mathbb{E}[\mathbf{R}^e] = -Cov(M_{APT}, \mathbf{R}^e) + \mathbb{E}[M_{APT}\mathbf{R}^e] . \quad (21)$$

These excess returns are the sum of a risk premium component, generated by their covariance with APT S-SDF risk, and a second component $\boldsymbol{\eta} = \mathbb{E}[M_{APT}\mathbf{R}^e]$ that satisfies the APT bound (16). In this sense, decomposition (21) is a model-free reproduction of the APT asset pricing predictions using APT S-SDFs. Such decomposition is naturally parametrized by threshold parameter τ . This parameter bounds the contribution of component $\boldsymbol{\eta}$ to expected excess returns in the arbitrage-free founding economy with ambiguity from Corollary 2.

Remark 7. Combining APT bound (19) and identity (21) with APT factor model (15) it follows:

$$\mathbb{E}[\mathbf{R}^e - \boldsymbol{\Lambda}\mathbf{F}^e]' \boldsymbol{\Sigma}_\zeta^{-1} \mathbb{E}[\mathbf{R}^e - \boldsymbol{\Lambda}\mathbf{F}^e] \leq \tau . \quad (22)$$

Under the assumption of a Gaussian residuals distribution with common covariance matrix $\boldsymbol{\Sigma}_\zeta$, this bound is equivalent to a relative entropy bound, which constrains the ambiguity between residual distributions with non-zero mean and a zero-mean residual distribution in factor model (15); see, e.g., Hansen and Sargent [2007].

Remark 8. Corollary 2 involves a closed-form penalization $\sigma_C(\boldsymbol{\theta}) = \tau \|\boldsymbol{\theta}_D\|_{2, \boldsymbol{\Sigma}_\zeta^{-1/2}}$, which founds a strictly positive S-SDF satisfying the APT pricing constraints (19) under pricing metric $\|\cdot\|_{2, \boldsymbol{\Sigma}_\zeta^{-1/2}}$. Clearly, these pricing constraints are satisfied also by any S-SDF M such that:

$$\mathbb{E}[M\mathbf{F}^e] = \mathbf{0} \text{ and } \|\mathbb{E}[M\mathbf{R}^e]\|_{\boldsymbol{\Sigma}_\zeta^{-1/2}} \leq \tau , \quad (23)$$

under a pricing metric such that $\|\cdot\| \geq \|\cdot\|_2$. Such a APT-SDF is founded in an arbitrage-

free economy with cost function $\sigma_C(\boldsymbol{\theta}) = \tau \|\boldsymbol{\theta}_D\|_{*, \Sigma_\zeta^{-1/2}}$. However, since $\|\cdot\| \geq \|\cdot\|_2$ the no arbitrage condition in such economy is more restrictive than the one adopted in Corollary 2. Therefore, while model-free APT S-SDF predictions are reproducible using a whole family of S-SDFs with distinct valuation geometries, the foundation of the APT S-SDFs in Corollary 2 relies on the least restrictive no-arbitrage condition.

4.3 Minimum dispersion APT-SDFs

APT S-SDFs are naturally suited for an analysis and selection framework based on minimum dispersion S-SDFs, in which we can coherently isolate and study the effects of distinct assumptions underlying APT asset pricing predictions, such as assumptions regarding the set of traded factors or the tightness of APT pricing error bound (16). This framework is made explicit by the next corollary of Proposition 2, which provides the basis for our empirical analysis in Section 5.

Corollary 3 (Minimum Dispersion APT S-SDF). *Given the above APT setting, consider the minimum dispersion APT S-SDF, $M_{0,APT}$, solving minimum dispersion problem (10) for constraint set $C = \{\mathbf{0}_{N_S+1}\} \times \{\boldsymbol{\eta} \in \mathbb{R}^{N_D} : \|\boldsymbol{\eta}\|_{\Sigma_\zeta^{-1/2}} \leq \tau\}$ under a norm $\|\cdot\| \geq \|\cdot\|_2$. Then, dual portfolio problem (11) reads in closed-form:*

$$\Delta(C) = \min_{\boldsymbol{\theta} \in \mathbb{R}^N} \left\{ \mathbb{E}[\phi_+^*(-\mathbf{X}'\boldsymbol{\theta})] + \theta + \tau \|\boldsymbol{\theta}_D\|_{*, \Sigma_\zeta^{-1/2}} \right\}. \quad (24)$$

Let further $\tilde{C} \subset \{\mathbf{0}_{N_S+1}\} \times \{\boldsymbol{\eta} \in \mathbb{R}^{N_D} : \|\boldsymbol{\eta}\|_{2, \Sigma_\zeta^{-1/2}} < \tau\}$ be closed, convex and contain $\mathbf{0}$, and the corresponding economy $(\mathcal{Z}_{\tilde{C}}, \pi_{\tilde{C}}, \mathbb{P})$ from Section 2.2 be arbitrage-free. Then, $\Pi(C) = -\Delta(C)$. If additionally $\boldsymbol{\theta}_0$ is a solution of problem (24) such that $-\mathbf{X}'\boldsymbol{\theta}_0 < \lim_{y \rightarrow \infty} \phi(y)/y$, $\mathbb{P}$ -almost surely, then either minimum S-SDF dispersion problem (10) has no solution, or it has a unique solution given by:

$$M_0 = (\phi_+^*)'(-\mathbf{X}'\boldsymbol{\theta}_0). \quad (25)$$

In summary, under appropriate conditions, minimum dispersion APT S–SDFs exist and can be computed as closed-form transformations of optimal portfolio payoffs from closed-form penalized portfolio problem (24).²⁵ In such problem, different choices of norm $\|\cdot\|$ imply potentially very different S–SDF properties, in terms of, e.g., S–SDF sparsity, which are however all compatible with the model-free APT predictions (19).

Remark 9. By construction, minimum dispersion APT S–SDF problems give rise to an extended family of closed-form SDF dispersion bounds, relative to the bounds in Hansen and Jagannathan [1991], Luttmer [1996] and Almeida and Garcia [2016], among others, which are parametrized by the APT pricing constraints. Given the choice of a norm $\|\cdot\|$, this reduces to a simple parametrization with threshold τ in APT pricing constraint (23). Therefore, let M_0^S be the unique S–SDF solving problem $\Pi(\infty) := \sup_{\tau > 0} \Pi(\tau)$, i.e., M_0^S is only required to have expectation 1 and to price the traded factor excess returns exactly, and define:

$$\tau^{max} := \|\mathbb{E}[M_0^S \mathbf{R}^e]\| . \quad (26)$$

τ^{max} defines a pricing constraint on excess returns satisfied by a normalized minimum dispersion APT SDF only required to price exactly the traded factors. The resulting generalized dispersion bound $\Pi(\tau)$ is strictly convex and strictly decreasing for any $\tau \leq \tau^{max}$, while it is constant at $\Pi(\tau) = E[\phi(M_0^S)]$ if $\tau \geq \tau^{max}$.²⁶

5 Empirics of APT S–SDFs

Starting from our theoretical analysis and selection framework for minimum dispersion APT S–SDFs, we empirically investigate the in-sample and out-of-sample pricing properties of a convenient family of APT S–SDFs, which differ with respect to their cross-sectional

²⁵See again Remark 3 for a more detailed discussion of the technical condition $-\mathbf{X}'\boldsymbol{\theta}_0 < \lim_{y \rightarrow \infty} \phi(y)/y$, $\mathbb{P}$ –almost surely.

²⁶This follows by [Luenberger, 1997, Prop. 1 and 2, p. 216-217], the strict convexity of ϕ –dispersions and the convexity of norms.

pricing accuracy and their degree of comovement with systematic asset return risks. We show that a tradeoff between these two key S-SDF properties naturally emerges, thus providing a criterion to compare APT S-SDFs corresponding to different pricing metrics, dispersion measures and choices of traded factors.

5.1 Data

The results in this section are based on standard datasets of characteristics sorted portfolio returns of varying cross-sectional dimensions, which build on monthly returns of the risk-free asset, the S&P 500 index as a proxy for the market index, the SMB and the HML factors and on the following selections of the double sorted portfolios from Kenneth French's data library:

1. **Low dimensional dataset:** 25 portfolio returns sorted on size and book to market, 10 portfolio returns sorted on momentum, and 25 portfolio returns sorted on size and long term reversal.
2. **Intermediate dimensional dataset:** 100 portfolio returns sorted on size and book to market, 25 portfolio returns sorted on momentum, 25 portfolio returns sorted on size and long term reversal, 25 portfolio returns sorted on size and short term reversal, and 49 portfolio returns sorted on industry.
3. **High dimensional dataset:** The portfolios in the intermediate dimensional dataset augmented with the characteristics-based factors from the WRDS financial ratios (WFR) dataset, which consists of 69 ratios for 10 industries based on Fama-French industry classification. The construction of the factor returns follows [Kozak et al. \[2020\]](#).

We also consider the returns of various Fama-French factors for constructing simple benchmark linear SDFs and for incorporating well-known systematic risk exposures as traded factors in our APT S-SDFs. Our sample starts in January 1931 (January 1970 for the

characteristic-based factors in the high dimensional dataset) and ends in June 2018. Portfolios with missing time series observations are removed. After such removal, the low dimensional dataset consists of 57 assets and 1054 time series observations, the intermediate dimensional dataset consists of 188 assets and 1054 time series observations, while the high dimensional dataset has 260 asset returns with 561 time series observations.

5.2 Empirical setting

In order to study APT S–SDFs with varying asset pricing and sparsity features, we consider for $\lambda \in [0, 1]$ the following APT norms:

$$h_{1, \Sigma_{\zeta}^{-1/2}}(\boldsymbol{\eta}_D) := (1 - \lambda) \|\boldsymbol{\eta}_D\|_{1, \Sigma_{\zeta}^{-1/2}} + \lambda \|\boldsymbol{\eta}_D\|_{2, \Sigma_{\zeta}^{-1/2}} , \quad (27)$$

and

$$h_{\infty, \Sigma_{\zeta}^{-1/2}}(\boldsymbol{\eta}_D) := (1 - \lambda) \sqrt{N_D} \|\boldsymbol{\eta}_D\|_{\infty, \Sigma_{\zeta}^{-1/2}} + \lambda \|\boldsymbol{\eta}_D\|_{2, \Sigma_{\zeta}^{-1/2}} . \quad (28)$$

Since $\|\cdot\|_2 \leq \|\cdot\|_1$ and $\|\cdot\|_2 \leq \sqrt{N_D} \|\cdot\|_{\infty}$, pricing constraints (23) based on these norms imply the benchmark APT pricing constraints (19). Moreover, while norm (27) induces varying degrees of sparsity on the resulting S–SDF valuations of dubious excess returns, norm (28) implies sparsity on the portfolio weights supporting minimum dispersion APT S–SDFs.²⁷

Estimators of S–SDF dispersion bounds and dual portfolio weights in stationary dynamic economies can be naturally developed. Let $\{\mathbf{X}_{t+1}\}_{t \in \mathbb{N}}$ be a time series of payoffs of the N basis assets, where $\mathbf{X}_{t+1} = (\mathbf{F}_{t+1}^e, \mathbf{R}_{t+1}^e)'$, defined on probability space $(\Omega, \mathcal{F}, \mathbb{P})$. This induces the following convex pricing constraints under the stationary probability $\mathbb{P}$:

$$\mathbb{E}[M_{t+1} \mathbf{F}_{t+1}^e] = \mathbf{0} \text{ and } \mathbb{E}[M_{t+1} \mathbf{R}_{t+1}^e] \in C . \quad (29)$$

²⁷These features follow from the presence of the l_1 –norm in pricing norm (27) and in the dual penalization $\sigma_{C_{\infty}} = \tau h_{\infty, \Sigma_{\zeta}^{-1/2}}$ from Corollary 3, respectively. See again Table 1 and Lemma OA-1 in Online Appendix A, which reports the closed-form expressions for penalization σ_{C_i} ($i = 1, \infty$) from Corollary 3.

By Corollary 3, the minimum S–SDF dispersion bound associated with convex pricing constraints of the form (29) equals $-\Delta(C)$. Given a sample of size $T > 0$, the estimator for population value $\Delta(C)$ is thus:

$$\Delta_T(C) := \min \{Q_T(\boldsymbol{\theta}) : \boldsymbol{\theta} \in \mathbb{R}^N\}, \quad (30)$$

with the empirical objective function

$$Q_T(\boldsymbol{\theta}) := \frac{1}{T} \sum_{t=1}^T [\phi_+^*(-\mathbf{X}'_{t+1}\boldsymbol{\theta})] + \theta + \sigma_C(\boldsymbol{\theta}).$$

Accordingly, $\boldsymbol{\theta}_T \in \arg \min_{\boldsymbol{\theta} \in \mathbb{R}^N} \{Q_T(\boldsymbol{\theta})\}$ is the corresponding estimator of dual optimal portfolio weight vector $\boldsymbol{\theta}_0 := \arg \min_{\boldsymbol{\theta} \in \mathbb{R}^N} \{\mathbb{E}[\phi_+^*(-\mathbf{X}'_{t+1}\boldsymbol{\theta})] + \theta + \sigma_C(\boldsymbol{\theta})\}$. We apply these estimators to compute minimum dispersion APT S–SDFs from Corollary 3 for pricing norms (27)–(28) and a set of measures of ϕ –dispersion from the Cressie and Read [1984] family; see again Online Appendix B.1.²⁸

5.3 Existence of empirical minimum dispersion S–SDFs

In Corollary 3, a dual APT S–SDF optimal portfolio representation (12) holds under a no arbitrage assumption for the underlying asset market $(\mathcal{Z}_{\tilde{C}}, \pi_{\tilde{C}}, \mathbb{P})$. Consequently, a duality failure ($\Pi(C) \neq -\Delta(C)$) is interpretable as a weak arbitrage-free foundation of a minimum dispersion APT S–SDF. Empirically, we find that a duality failure can arise for constraint sets C associated with low thresholds τ . Figure 1 documents this for the high dimensional dataset. Here, we estimate various minimum variance APT S–SDFs via their empirical dual portfolio problems. We then compute the empirical quantity $\left\| \frac{1}{T} \sum_{t=1}^T (M_{0t+1} \mathbf{X}_{t+1}) \right\|_{2, \boldsymbol{\Sigma}^{-1/2}}$ and verify whether it is smaller than the imposed threshold τ .

²⁸While a formal study of the asymptotic properties of estimator $\boldsymbol{\theta}_T$ is beyond the scope of this paper, note that $\boldsymbol{\theta}_T$ is an M –estimator with a strictly convex but possibly nonsmooth objective function, when penalty σ_C is not differentiable. Asymptotic properties of such estimators are established under a fix dimension N , e.g., with the techniques in Andrews [1994]. Negahban et al. [2012] offer a unified framework for establishing consistency and convergence rates of high-dimensional M –estimators with norm based penalizations.

We find that for thresholds smaller than discontinuity point $\hat{\tau}^{min}$ in Figure 1 an empirical duality failure emerges, i.e., no strictly positive minimum variance empirical APT SDF exists. By recognizing the possibility of non-existence of a strictly positive empirical APT S–SDF, we can discipline economically the choice of penalization parameter τ .²⁹ In our empirical analysis, penalization parameters are constrained to be not smaller than lower bound $\hat{\tau}^{min}$. Moreover, as threshold τ increases, the pricing constraint on minimum dispersion APT S–SDFs is progressively relaxed, to the point that a minimum dispersion APT S–SDF only required to exactly price the traded factors satisfies the constraints for a sufficiently large threshold $\hat{\tau}^{max}$; see again Equation (26). In our empirical analysis, we ensure a well-defined empirical tradeoff between APT S–SDF spanning features for systematic risks and their pricing accuracy by constraining admissible penalization parameters in the interval $[\hat{\tau}^{min}, \hat{\tau}^{max}]$.

5.4 In-sample analysis

In the in-sample analysis, we characterize in detail the properties of the attainable tradeoffs between pricing accuracy and spanning properties for systematic return risks generated by APT S–SDFs under varying assumptions on the latter.

5.4.1 Tradeoff between cross-sectional and time series S–SDF explanatory power

We empirically explore the tradeoffs between pricing accuracy and spanning properties for systematic return risks of minimum variance APT S–SDFs.³⁰ In this section, we take

²⁹In principle, an empirical duality failure may arise because of the ill-posedness of the empirical minimum S–SDF dispersion problem, also when when no duality failure may appear at the population level. We thank an anonymous referee for pointing this out.

³⁰We focus on minimum variance APT S–SDFs, because as explained below and documented empirically in Online Appendix C.1 they always outperform by construction other minimum dispersion APT S–SDFs in terms of such tradeoffs. This makes variance the natural measure of dispersion to optimize these tradeoffs.

the market excess return as the single traded factor. Therefore, these APT S–SDFs are interpretable as minimum variance corrections of an empirical SDF under the CAPM, which allow for bounded misvaluations of excess returns orthogonal to market risk. While in Section 5.4.3 we also explore the implications of the choice of traded factors for the resulting S–SDF tradeoffs, one may argue that tradability of market returns is the most robust economically motivated assumption one may rely on when assuming tradability of some test assets. Moreover, since all theoretical asset pricing predictions in Section 4 hold independent of whether further traded factors beyond the assumed ones exist, the resulting APT S–SDF pricing predictions are similarly robust to the presence of further traded factors. Figure 2 quantifies the tradeoffs in the low dimensional dataset, based on different norms (27) and (28) such that $\lambda \in \{0, 0.5, 1\}$. Note that when $\lambda = 1$ norm $h_1 = \|\cdot\|_1$ ($h_\infty = \sqrt{N_D} \|\cdot\|_\infty$) implies extremely sparse valuation errors (S–SDFs dual portfolios), while for $\lambda = 0$ no sparsity in valuations errors and S–SDFs arises, because $h_1 = h_\infty = \|\cdot\|_2$. Parameter value $\lambda = 0.5$ implies instead a less extreme degree of sparsity in misvaluations (norm h_1) or S–SDFs (norm h_∞).³¹

Panel (A) of Figure 2 summarizes the tradeoff between minimum S–SDF volatility and pricing threshold τ , in an extended version of the minimum S–SDF volatility bounds in Hansen and Jagannathan [1991] and Luttmer [1996]. Since different norms give rise to a different maximal threshold $\hat{\tau}^{max}$ implied by the S–SDF only required to price exactly market excess returns, we collect minimum S–SDF volatilities parametrized by relative thresholds $\kappa := \tau / \hat{\tau}^{max} \in [\hat{\tau}^{min} / \hat{\tau}^{max}, 1]$.

In the right plot of Panel (B) of Figure 2, the minimum S–SDF volatilities induced by pricing constraints under the APT l_2 –norm (l_∞ –norm) correspond to the highest (lowest) S–SDF cross-sectional explanatory power, when the explanatory power is measured by the

³¹Section 5.5 documents these sparsity properties in more detail.

standard cross-sectional GLS R^2 metric:³²

$$R_{GLS}^2 := 1 - \frac{\|\mathbb{E}[M_0(\mathbf{X} - \bar{\mathbf{X}}\mathbf{1})]\|_{2, \Sigma^{-1/2}}^2}{\|\mathbb{E}[\mathbf{X} - \bar{\mathbf{X}}\mathbf{1}]\|_{2, \Sigma^{-1/2}}^2}, \quad (31)$$

where $\mathbf{1}$ is a $N \times 1$ vector of ones, $\bar{\mathbf{X}} := (\mathbf{1}'\mathbf{X})/N$ the average of the components of vector $\mathbf{X}$ and Σ the $N \times N$ covariance matrix of excess return vector $\mathbf{X}$. This correspondence offers a second useful interpretation of penalization parameter τ in Corollary 3, which can be understood as a parametrization of the relative loss $1 - R_{GLS}^2$ in cross-sectional explanatory power that an APT S-SDF suffers with respect to the minimum dispersion SDF exactly pricing all assets. Because of the scale independence of R_{GLS}^2 , this interpretation simplifies the comparison of minimum dispersion APT S-SDFs induced by different pricing constraints.³³

Given a set of traded systematic factor excess returns, the minimum S-SDF volatilities induced by APT pricing constraints are also directly related to the S-SDF ability to comove with systematic excess return risks. Indeed, since traded excess factor returns are priced exactly, lower S-SDF pricing errors are linked to both larger S-SDF volatilities and lower absolute correlations with traded systematic risks. This constraint impacts the time series S-SDF comovement with any excess return X , as measured by a standard time series OLS R^2 :

$$R_{OLS}^2 := \frac{(Cov(M_0, X))^2}{Var(M_0)Var(X)} = \frac{1}{Var(M_0)} \cdot \frac{(\mathbb{E}[X] - \mathbb{E}[M_0X])^2}{Var(X)}. \quad (32)$$

Given a cross-section of asset (squared) Sharpe ratios, identity (32) constraints the S-SDF average time series explanatory power in the left plot of Panel (B) from Figure 2, as a function of the varying relative threshold κ . A higher cross-sectional explanatory power in the bottom right plot corresponds to a lower time series explanatory power in the bottom

³²By definition, R_{GLS}^2 equals one minus the ratio of the cross-sectional variance of pricing errors and the cross-sectional variance of expected returns, after a standardization by the return volatility matrix $\Sigma^{1/2}$.

³³As shown in Lewellen et al. [2010], R_{GLS}^2 is a more appropriate measure of cross-sectional fit when pricing cross-sections of sorted portfolio returns.

left plot. However, what matters for an economic S–SDF comparison is the generated relative tradeoff between time series and cross-sectional explanatory powers. Panel (B) of Figure 2 shows that minimum variance S–SDFs under the APT l_2 –norm always imply the most favourable tradeoff in-sample. For instance, while under this norm a cross-sectional GLS R^2 of 50% can be obtained together with an average time series R^2 of about 28%, under an APT l_1 – and l_∞ –norm the resulting average time series R^2 is only about 20% and 14%, respectively. This evidence naturally follows from the fact that the l_2 –pricing constraint is the only one producing for any given target value of cross-sectional GLS R^2 metric (31) the corresponding minimum variance S–SDF. Given the moment structure of excess returns, minimizing the S–SDF variance is equivalent to maximizing the time-series R^2 metric (32) with respect to the traded systematic risk factors. Therefore, the minimum variance S–SDF with l_2 –pricing constraint produces a Pareto improved tradeoff under the standard GLS R^2 metric (32) of cross-sectional explanatory power.³⁴

5.4.2 Metrics of pricing accuracy and robustness of S–SDF tradeoff

Two further important aspects for understanding the S–SDF tradeoff between pricing accuracy and time series explanatory power cover: (i) the role of the metric of pricing accuracy and (ii) the robustness of the tradeoff in presence of deviations from the in-sample assumptions.³⁵ Regarding the former aspect, recall that APT S–SDFs satisfy APT pricing constraints (23), under a norm $\|\cdot\|$ that may be different from the l_2 –norm. Therefore, the natural equivalent of GLS R^2 metric (31) for measuring the S–SDF cross-sectional

³⁴Figure OA-3 in the Online Appendix shows that absolute S–SDF betas monotonically decrease with threshold τ , as intuitively expected. Lewellen et al. [2010] emphasize the challenges arising in weakly identified empirical asset pricing settings relying on test assets with a hidden factor structure. In this context, incorporating well-chosen traded systematic risks in our framework can be interpreted as an informal S–SDF regularization against weak factor features.

³⁵Intuitively, the second aspect relates to the fact that empirical S–SDFs with better resistance to varying in-sample data features are likely to perform better out-of-sample.

explanatory power is:

$$R_{GLS, \|\cdot\|}^2 := 1 - \frac{\|\mathbb{E}[M_0(\mathbf{X} - \bar{\mathbf{X}}\mathbf{1})]\|_{\Sigma^{-1/2}}^2}{\|\mathbb{E}[\mathbf{X} - \bar{\mathbf{X}}\mathbf{1}]\|_{\Sigma^{-1/2}}^2}. \quad (33)$$

A natural question is how the tradeoff in Section 5.4.1 between cross-sectional and time series explanatory power is impacted by the choice of the metric of pricing accuracy. For each minimum variance APT S-SDF in Panel (B) of Figure 2, Figure 3 plots this tradeoff using cross-sectional GLS R^2 metric (33) with norms $\|\cdot\|_1$, $\|\cdot\|_2$, and $\sqrt{N_D}\|\cdot\|_\infty$. The three left plots show that minimum variance APT S-SDFs produce a Pareto dominating tradeoff when metric (33) is defined by the same norm as the one in APT bound (23). At the same time, it appears that the tradeoff induced by APT S-SDFs with l_2 -pricing constraint is less sensitive to the choice of the GLS R^2 metric than the tradeoffs induced by l_1 - and (scaled) l_∞ -pricing constraints.

Concerning the second aspect above, the three right plots of Figure 3 illustrate the effects of data perturbations, creating a discrepancy of about 10 years of monthly observations between S-SDF estimation and evaluation samples, for the S-SDF tradeoff between time series and cross-sectional explanatory powers. We find that APT S-SDFs satisfying an l_2 -pricing constraint tend to produce a more consistent tradeoff on perturbed data, essentially outperforming the other S-SDFs with respect to all GLS R^2 metrics.

5.4.3 Traded factors and systematic S-SDF risks

Intuitively, traded factors in our framework correspond to tradeable systematic excess returns with a clearly understood risk compensation. As mentioned, the exact pricing condition on traded factor excess returns forces via equation (32) a maximal squared correlation with minimum variance S-SDFs, i.e., these S-SDFs maximize the comovement with the systematic risks spanned by the traded factors. Hence, the choice of the traded factor returns crucially influences the S-SDF tradeoff between cross-sectional and time

series explanatory power.

Figure 4 explores these tradeoffs for various minimum variance S-SDFs with a l_2 -APT pricing constraint (16), under different choices of the traded factors: (i) no traded factor, (ii) one single traded factor given by the market excess return or by the first principal component of excess returns, and (iii) three traded factors given by the three Fama-French factor excess returns. We find that all S-SDFs imply a virtually identical cross-sectional GLS R^2 curve in-sample.³⁶ However, these S-SDFs in part differ in the way how they span systematic excess return risks. The S-SDFs including the market excess return or the first principal component of excess returns as a single traded factor produce similar and highest average time series R^2 curves. In contrast, S-SDFs including no traded factor uniformly produce a very low time series explanatory power that is essentially independent of the chosen degree of penalization. Therefore, including tradeable risk exposures that reflect well-understood systematic risks allows to balance the S-SDF trade-off between time series and cross-sectional explanatory power of APT S-SDFs under metrics (31) and (32). Extending the set of traded factors to include the three Fama-French factor returns produces a minimum variance correction of an empirical SDF under the three-factor Fama-French model. Interestingly, we find that this extension gives rise to a tradeoff that is Pareto dominated by the one of the simpler minimum variance correction of a CAPM SDF.

Overall, we conclude that minimum variance APT S-SDFs constraining valuations of asset excess returns with APT bound (19) and exactly pricing market excess returns or the first principal component of sorted excess returns consistently offer the best in-sample tradeoff between pricing accuracy and time series explanatory power.

³⁶With the only distinction that in the intermediate dimensional dataset the empirical minimum variance S-SDF including the Fama-French factors as traded factors does not exist for a sufficiently low threshold τ .

5.5 The role of sparsity in APT S–SDFs

As emphasized in Section 3.1, sparsity cannot be obtained at the same time for the misvaluations of excess returns and the dual optimal portfolios of a minimum dispersion S–SDFs. Therefore, in an APT setting one can either impose sparsity on scaled vector $\Sigma_{\zeta}^{-1/2}\mathbb{E}[M_0\mathbf{X}_D]$ or on scaled portfolio weights $\Sigma_{\zeta}^{1/2}\boldsymbol{\theta}_{0D}$. Figure 5 documents these sparsity properties for the three minimum variance S–SDFs with 50% target GLS R^2 in Panel (B) of Figure 2.

In Panel (A) of Figure 5, the l_1 –pricing norm gives rise to sparse excess return misvaluations. In contrast, the (scaled) l_{∞} –norm produces extreme shrinkage and no sparsity. The l_2 –norm induces shrinkage and no sparsity as well, but with maximal excess return misvaluations larger than under the (scaled) l_{∞} –norm. The properties of S–SDF dual portfolio weights follow from the corresponding penalizations $\sigma_C = \tau \|\cdot\|_{*,\Sigma_{\zeta}}$ in Corollary 3. Since the dual norm of the l_1 –norm is the l_{∞} –norm, the l_1 –pricing constraint produces extreme shrinkage in dual portfolio weights and no sparsity. Analogously, the (scaled) l_{∞} –norm gives rise with a l_1 –penalization to extreme sparsity in dual portfolio weights. Finally, the self-duality of the l_2 –norm induces shrinkage but no sparsity in dual portfolio weights.

While linked to S–SDFs with different sparsity properties, the S–SDF time series in Panel (B) of Figure 5 are superficially quite similar in a number of dimensions.³⁷ However, they give rise to economically important differences in time series explanatory power for asset returns. As a consequence, the choice of a norm in the S–SDF pricing constraints and the corresponding S–SDF sparsity features are in the end strongly linked to a particular tradeoff between S–SDF cross-sectional and time series explanatory power. Our minimum dispersion APT S–SDF theory in Corollary 3 makes this tradeoff empirically measurable and indicates that S–SDF sparsity is quite costly with respect to the implied tradeoff between cross-sectional and time series explanatory powers under metrics (31) and (32).

³⁷All S–SDF time series exhibit quite similar volatilities between about 1.75 and 1.85, as well as a quite substantial co-movement, with time series correlations above 0.8 between S–SDFs induced by the l_2 –norm and the other S–SDFs.

5.6 APT S–SDFs, Non APT S–SDFs and Relation to Kozak et al. [2020]

When pricing accuracy is measured by the standard, not self-standardized, cross-sectional R^2 metric:

$$R_{OLS}^2 := 1 - \frac{\|\mathbb{E}[M_0(\mathbf{X} - \bar{\mathbf{X}}\mathbf{1})]\|_2^2}{\|\mathbb{E}[\mathbf{X} - \bar{\mathbf{X}}\mathbf{1}]\|_2^2}, \quad (34)$$

the corresponding optimal minimum variance S–SDFs simply need to restrict the pricing with a constrained form of the l_2 -pricing constraint (19), in which $\Sigma := \mathbf{I}_{N \times N}$, i.e., such optimal S–SDFs are by construction non APT S–SDFs.

The two top panels of Figure 6 illustrate for the high dimensional dataset the optimal tradeoffs implied by minimum variance APT and non APT S–SDFs exactly pricing the first principal component of excess returns, under metrics of pricing accuracy (31) and metric (34), respectively.³⁸ Importantly, both such S–SDFs are sparse in two corresponding portfolios of excess returns: the first principal component of excess returns itself and a second optimal portfolio of returns orthogonal to traded systematic risk, which bounds the overall S–SDF mispricing with an APT bound (31) and with its constrained non APT version such that $\Sigma_\zeta := \mathbf{I}_{N_D \times N_D}$, respectively.

Kozak et al. [2020] study the sparsity implications of minimum variance S–SDFs in large asset markets. As elaborated in Section 3, their S–SDFs can be obtained with an Elastic Net norm penalization $\sigma_C = \alpha \|\cdot\|_1 + \tau \|\cdot\|_2$; see again Table 1. For $\alpha = 0$ these S–SDFs are not sparse and optimize the cross-sectional pricing accuracy under metric (34). In contrast, for $\alpha \neq 0$ they are sparse and give rise to a suboptimal tradeoff between the S–SDF time series explanatory power and pricing accuracy under metric (34). However, Kozak et al. [2020] show that this sparsity-induced suboptimality is less serious when it is imposed on the set of all principal components of returns, rather than on the original returns.

The bottom right panel of Figure 6 illustrates the degree of suboptimality of the tradeoff

³⁸Analogous evidence for the low dimensional dataset is reported in Figure OA-2 of the Online Appendix.

between S-SDF time series explanatory power and pricing accuracy under metric (34), when sparsity is induced on the endogenously selected principal components in Kozak et al. [2020] S-SDFs. It additionally shows that the tradeoff of a non APT S-SDF exactly pricing the first principal component of excess returns alone is essentially identical to the one of the optimal non sparse Kozak et al. [2020] S-SDF, which depends on all principal components of excess returns. Finally, the bottom left panel of Figure 6 shows that principal component APT S-SDFs not forcing exact pricing of the first principal component are on average essentially uncorrelated with excess returns, i.e., they are unable to generate a tradeoff between S-SDF time series explanatory power and pricing accuracy.

In summary, our minimum variance APT and non APT S-SDFs exactly pricing the first principal component of excess returns provide an optimal tradeoff between pricing accuracy and time series explanatory power under metrics (31) and (34) of cross-sectional pricing accuracy. These S-SDFs are by construction sparse in the returns of two distinct portfolios, given by the first principal component of returns and a second optimal portfolio that bounds the mispricing across assets under l_2 -APT pricing constraint (19) and under l_2 -non APT pricing constraint (19) with $\Sigma_\zeta := \mathbf{I}_{N_D \times N_D}$, respectively.

5.7 Out-of-sample analysis

We next characterize the out-of-sample tradeoff between pricing accuracy and time series explanatory power generated by minimum variance APT S-SDFs.³⁹ To this end, we estimate data-driven out-of-sample minimum variance APT S-SDFs in a non forward-looking manner.⁴⁰ We then evaluate the out-of-sample S-SDF tradeoff between pricing accuracy and time series explanatory power according to metrics (31) and (32). Given our earlier results, we focus on minimum variance APT S-SDFs implying varying sparsity proper-

³⁹We adopt a simple approach not requiring an explicit modelling of conditioning information. While the modelling of conditioning information is an important dimension we plan to address in future research, it would require a more extended treatment formally dealing with a possibly time varying constraint set C , a potentially large number of information variables, and a growing cross-sectional asset dimension.

⁴⁰A forward-looking out-of-sample analysis, further supporting the evidence presented in this section, can be found in Online Appendix C.2.

ties under pricing constraints with norms $\|\cdot\|_1$, $\|\cdot\|_2$, and $\sqrt{N_D} \|\cdot\|_\infty$, and consider various choices for the set of traded factor returns, such as the market excess return alone.

To implement optimal minimum variance APT S-SDFs in a non forward-looking way, we propose a data-driven approach, which incorporates varying views on the desired in-sample S-SDF tradeoff between times series and cross-sectional explanatory power. These views may implicitly incorporate, e.g., the importance that a researcher/model builder assigns to these two tradeoff dimensions. For this purpose, we consider a sequence of estimation windows of 30 years updated at a semi-annual frequency, where in each 30-year window the optimal S-SDF dual portfolio weights are estimated and used to construct the out of sample S-SDFs for the 6 months following the estimation window. Specifically, given an estimation window of 30 years with last observation in month y , the vector of optimal portfolio weights, $\hat{\theta}_y^*$, is estimated by the solution of empirical optimization problem (30) for a corresponding optimal choice of the pricing threshold in the estimation window. The out-of-sample S-SDFs are then computed as $\hat{M}_{y+m}^* = \max\{-\mathbf{X}'_{y+m}\hat{\theta}_y^*, 0\}$ for $m = 1, \dots, 6$.⁴¹ By rolling the 30-year window at a semi-annual frequency we obtain a monthly time series of data-driven APT S-SDFs for which we can evaluate the out-of-sample time series and cross-sectional explanatory power.

In order to select the optimal pricing threshold, we split each 30-year estimation window in a training window of 25 years and a separate window of 5 years. For a range of admissible relative pricing thresholds $\hat{\kappa}_y \in [\hat{\kappa}_y^{min}, 1]$ corresponding to the training window, we estimate various minimum variance APT S-SDFs using only the training data.⁴² Based on the entire window of 30 years of data, we then compute the resulting curves of S-SDF metrics $R_{GLS}^2(\hat{\kappa}_y)$ and $R^2(\hat{\kappa}_y)$ defined in equations (31) and (32). Finally, we select an optimal pricing threshold $\hat{\kappa}_y^* := \hat{\kappa}_y^*(p)$, by maximizing cross-sectional metric $R_{GLS}^2(\hat{\kappa}_y)$ under a

⁴¹The specific shape of $\hat{M}_{y+m}^*$ emerges from the closed-form link between minimum variance S-SDF and optimal dual portfolio payoff found in Corollary OA-2 of Online Appendix B.1.

⁴²Recall that the minimum admissible relative threshold is given by $\hat{\kappa}_y^{min} = \hat{\tau}_y^{min}/\hat{\tau}_y^{max}$, where $\hat{\kappa}_y^{min}$ is the smallest threshold value for which in the training window no empirical duality failure arises and $\hat{\tau}_y^{max}$ is given by the empirical version of equation (26).

constraint on the average time series metric $\overline{R^2}(\hat{\kappa}_y)$ across assets. This constraint requires $\overline{R^2}(\hat{\kappa}_y)$ to be not lower than a fraction $p \in [0, 1]$ of the average time series metric $\overline{R^2}(1)$ under an empirical SDF only pricing exactly the traded factor excess returns:

$$\hat{\kappa}_y^* = \arg \max_{\hat{\kappa}_y \in [\hat{\kappa}_y^{min}, 1]} \{R_{GLS}^2(\hat{\kappa}_y) : \overline{R^2}(\hat{\kappa}_y) \geq p\overline{R^2}(1)\} . \quad (35)$$

We compute optimal threshold $\hat{\kappa}_y^*$ for the choices $p \in \{0, 0.25, 0.5, 0.75, 1\}$, where the first choice corresponds to an optimization (35) with no constraint on the S-SDF time series explanatory power.

Figure 7 collects the out-of-sample evidence on data-driven minimum variance S-SDFs for various intuitive choices of the set of traded factor returns. Given our earlier results, we focus on l_2 -APT pricing constraints. In the top two panels of Figure 7, we obtain an out-of-sample tradeoff between S-SDF time series and cross-sectional explanatory power that is consistent with the in-sample selection of the sequence of optimal thresholds $\{\hat{\kappa}_y^*\}$. For instance, for the S-SDFs including the market excess return as the single traded factor, the out-of-sample GLS R^2 metric (31) in the low (intermediate) dimensional data set decreases monotonically with the tightness of the constraint on the in-sample time series explanatory power, yielding GLS R^2 s between a maximum of 68% (38%) and a minimum of -2% (-2%) for parameters $p \in \{0, 0.25, 0.5, 0.75, 1\}$. Moreover, in all cases the maximum cross-sectional GLS R^2 is attained by APT S-SDFs with tuning parameter $\hat{\kappa}_y^* > 0$, i.e., standard minimum variance SDFs produce a suboptimal cross-sectional fit. On the other hand, the out-of-sample average time series R^2 s are increasing with the tightness of the in-sample constraint from a minimum of 2.8% (1%) to a maximum of 68% (62%). An analogous pattern arises for other choices of traded factors, given by the first principal component of returns and the three Fama-French factors, respectively.

The parameter choice $p = 1$ corresponds to minimum variance S-SDFs only required to price exactly the traded factors in-sample. Given the choice of the market excess return as the single traded factor, the resulting S-SDF produces a mildly negative cross-sectional

GLS R^2 , together with a large average time series R^2 of 62% (68%) in the intermediate (low) dimensional dataset. In comparison, a minimum variance S-SDF only required to price exactly the three Fama-French factors produces an even more negative GLS R^2 and an average time series R^2 of only 18% (22%) in the intermediate (low) dimensional dataset. This explains why the minimum variance S-SDFs incorporating the three Fama-French factors as traded factors produce a Pareto dominated out-of-sample tradeoff between time series and cross-sectional explanatory power relative to the tradeoffs of S-SDFs incorporating the market alone as a traded factor.

A further Pareto improved tradeoff is achieved by minimum variance S-SDFs incorporating the first principal components of returns as a single traded factor in-sample. Interestingly, while these S-SDFs tend to produce larger average time series R^2 s, due to the more consistent out-of-sample time series explanatory power of the first principal component of returns relative to market returns, the resulting tradeoff between time series and cross-sectional explanatory power is quite comparable to the one induced by the S-SDF pricing exactly market returns alone. Notably, this S-SDF is a simple minimum variance correction of an empirical SDF under the CAPM, which exactly prices market risk but otherwise bounds the amount of mispricing across assets with a standard APT pricing constraint.

Following the intuition underlying the APT, model-free APT S-SDFs able to partly span systematic return shocks need to correlate with potential common risk factors in asset returns. The bottom two panels of Figure 7 illustrate this intuition more precisely, by reporting the fraction of out-of-sample variation in model-free APT S-SDFs that is explained by variations of the first principal component of sorted portfolios excess returns. Consistent with our previous findings, APT S-SDFs maximizing the in-sample pricing accuracy produce the highest out-of-sample pricing accuracy, but they are also almost uncorrelated with the first principal component of excess returns out-of-sample. Conversely, APT S-SDFs only required to exactly price systematic risk exposures in-sample produce the lowest

out-of-sample pricing accuracy and the largest comovement with the first principal component of excess returns. Such comovement is naturally highest, and rather similar, for S-SDFs exactly pricing either market returns or the first principal component of sorted portfolios excess returns in-sample.

Finally, we also find that the APT S-SDFs in Figure 7 clearly outperform linear SDF specifications in the traded factor excess returns that are computed by globally minimizing the in-sample pricing error size according to a standard GMM criterion. As for the APT S-SDFs of Figure 7, we estimate such SDFs on rolling windows of 30 years of monthly data, while updating the resulting GMM estimates at a semi-annual frequency. We then evaluate their out-of-sample cross-sectional and time series explanatory power for different choices of the set of factors. In all cases, we obtain pairs of cross-sectional and time series explanatory power that are Pareto dominated by corresponding pairs induced by some minimum variance S-SDF in Figure 7. For instance, a CAPM SDF specification with a single market factor implies a cross-sectional GLS R^2 of -0.2% (0.5%) and an average time series R^2 of 51.8% (52%) in the low (intermediate) dimensional dataset. Similarly a three-factor Fama-French SDF specification implies a cross-sectional GLS R^2 of 2.8% (0.4%) and an average time series R^2 of 26.9% (26.3%) in the low (intermediate) dimensional dataset.

6 Conclusions

This paper introduces a unifying framework for analyzing and selecting Stochastic Discount Factors (SDFs) in arbitrage-free markets with convex pricing constraints. These SDFs have an arbitrage-free foundation in economies with frictions or economies with ambiguity. In economies with frictions, they represent a positive linear pricing rule that bounds from below the nonlinear pricing rule in the market with frictions. In frictionless economies with ambiguity, they represent the positive linear pricing rule of an unobserved marginal investor, but under a different probability belief than the marginal investor's belief. This arbitrage-free framework provides an independent foundation for several important asset

pricing theories, such as Ross [1976]' Arbitrage Pricing Theory (APT), the good-deal bound SDF theory in Cochrane and Saa-Requejo [2000], various approaches to SDF regularization such as Kozak et al. [2020], and robust approaches for the identification of investors' beliefs such as Chen et al. [2020].

Our SDF analysis and selection framework is based on minimum dispersion SDF problems subject to convex pricing constraints, which naturally extend the classical minimum variance SDF methodology in Hansen and Jagannathan [1991]. We establish a duality between minimum dispersion SDF problems and penalized dual portfolio selection problems, in which the portfolio penalization stays in a one-to-one relation with the SDF convex pricing constraints. Here, minimum dispersion SDFs are identified as simple transformation of the dual optimal payoff of a penalized portfolio, which gives rise to a SDF regularization that is economically interpretable in terms of an unobserved structure of frictions or ambiguity in a corresponding arbitrage-free market.

We use market data to extract minimum variance APT SDFs that summarize with a model-free approach the admissible tradeoffs between cross-sectional pricing accuracy and SDF co-movement with systematic return risks. We find that a simple minimum variance correction of a CAPM-SDF, in which the mispricing of risks orthogonal to market risk is bounded according to APT pricing constraints, produces a Pareto optimal tradeoff. Such Pareto optimal SDF corrections are founded in arbitrage-free frictionless markets with ambiguity and they depend on two economically distinct SDF factors alone. The first factor is a traded excess return that maximally correlates with systematic return risks. The second factor consists of a minimum variance portfolio excess return, which optimally bounds the mispricing of risks unspanned by the traded SDF factor. Therefore, the Pareto optimal tradeoffs attained by our minimum variance SDF corrections correspond to sparse SDFs dependent on the excess returns of just two economically interpretable portfolios.

Appendix A - Tables and Figures

Table 1: Convex pricing constraints and supporting cost functions.

Pricing constraint and supporting cost function	h	σ_C
Conic	δ_K	δ_{K^o}
Norm	$\ \cdot\ $	$\tau \ \cdot\ _*$
Self-standardized norm	$\ \cdot\ _{\mathbf{W}^{-1}}$	$\tau \ \cdot\ _{*,\mathbf{W}}$
Convex combination of norms	$(1-\lambda) \ \cdot\ ^{(1)} + \lambda \ \cdot\ ^{(2)}$	$\tau \left(\min_{\mathbf{z} \in \mathbb{R}^N} \left\{ \max \left\{ \frac{\ \cdot\ _*^{(1)}}{1-\lambda}, \frac{\ \cdot - \mathbf{z}\ _*^{(2)}}{\lambda} \right\} \right\} \right)$
Euclidean distance from norm ball	$\text{dist}_{\alpha B(\ \cdot\)}$	$\alpha \ \cdot\ _* + \tau \ \cdot\ _2$
Short-sale constraint	$\delta_{\mathbb{R}_-^{N_D}}$	$\delta_{\mathbb{R}_+^N}$
l_2 -norm	$\ \cdot\ _2$	$\tau \ \cdot\ _2$
l_1 -norm	$\ \cdot\ _1$	$\tau \ \cdot\ _\infty$
Self-standardized l_∞ -norm	$\ \cdot\ _{\infty, \mathbf{W}^{-1}}$	$\tau \ \cdot\ _{1, \mathbf{W}}$
Convex combination of l_1 - and l_2 -norms	$(1-\lambda) \ \cdot\ _1 + \lambda \ \cdot\ _2$	$\tau \left(\min_{\mathbf{z} \in \mathbb{R}^N} \left\{ \max \left\{ \frac{\ \cdot\ _\infty}{1-\lambda}, \frac{\ \cdot - \mathbf{z}\ _2}{\lambda} \right\} \right\} \right)$
Distance from l_∞ -ball	$\text{dist}_{\alpha B_\infty}$	$\alpha \ \cdot\ _1 + \tau \ \cdot\ _2$

The table collects various specifications of function h parametrizing constraint set C , i.e., $C = \{\boldsymbol{\eta} \in \mathbb{R}^N : h(\boldsymbol{\eta}) \leq \tau\}$, with $\tau \geq 0$, and associated supporting cost functions σ_C , along with closed-form examples. From top to bottom, we consider: (i) conic pricing constraints obtained via characteristic functions of convex cone K and its polar cone K^o ; for instance, short-sale constraints feature $K = \mathbb{R}_-^N$ and $K^o = \mathbb{R}_+^N$ while bid-ask spreads feature $K = \mathbb{R}_-^N \times \mathbb{R}_+^N$ and $K^o = \mathbb{R}_+^N \times \mathbb{R}_-^N$. (ii) norm constraints obtained via $h = \|\cdot\|$ with $\sigma_C = \tau \|\cdot\|_*$, where $\|\cdot\|_*$ is the dual norm of $\|\cdot\|$; for instance, the l_2 -norm is self-dual, while the l_1 - and l_∞ -norms are dual to each other. (iii) self-standardized norms defined via $\|\cdot\|_{\mathbf{W}^{-1}} := \|\mathbf{W}^{-1} \cdot\|$. (iv) Euclidean distance from a ball of radius α under norm $\|\cdot\|$ (i.e., $\alpha B(\|\cdot\|) := \{\boldsymbol{\eta} \in \mathbb{R}^N : \|\boldsymbol{\eta}\| \leq \alpha\}$) defined via $\text{dist}_{\alpha B(\|\cdot\|)} := \inf_{\boldsymbol{\eta} \in \mathbb{R}^N} \{\|\cdot - \boldsymbol{\eta}\|_2 : \|\boldsymbol{\eta}\| \leq \alpha\}$. In the table, $\tau \geq 0$, $\lambda \in [0, 1]$ and $\mathbf{W}$ denotes a generic symmetric positive definite weighting matrix.

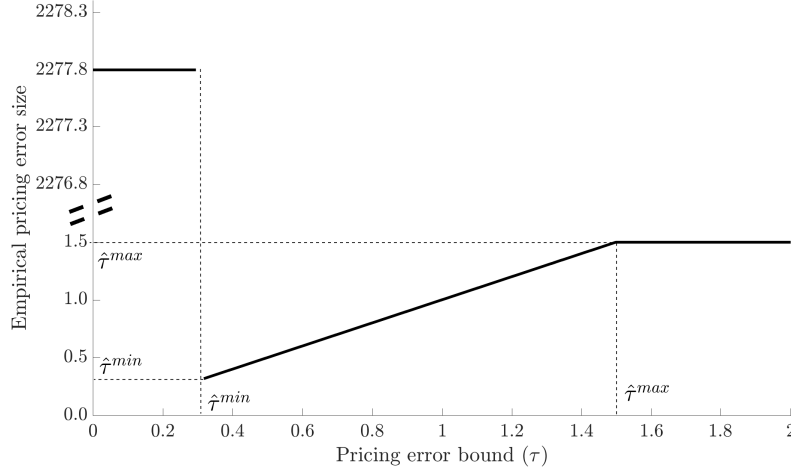


Figure 1: **Empirical duality failure.** We compute minimum variance S–SDFs satisfying standard APT pricing constraints. The figure reports the estimate of $\|\mathbb{E}[M_0 \mathbf{X} - \mathbf{P}]\|_{2, \Sigma^{-1/2}}$ for varying pricing thresholds τ . The point of discontinuity in the plot identifies the smallest threshold τ , for which a solution of the empirical primal S–SDF problem exists. The largest threshold $\hat{\tau}^{max}$ in the plot is computed as the sample version of the maximal threshold τ^{max} in equation (26). Sorted portfolio returns are used to construct the S–SDFs without assuming any observable traded factor. All calculations are based on the high dimensional dataset from January 1970 to June 2018.

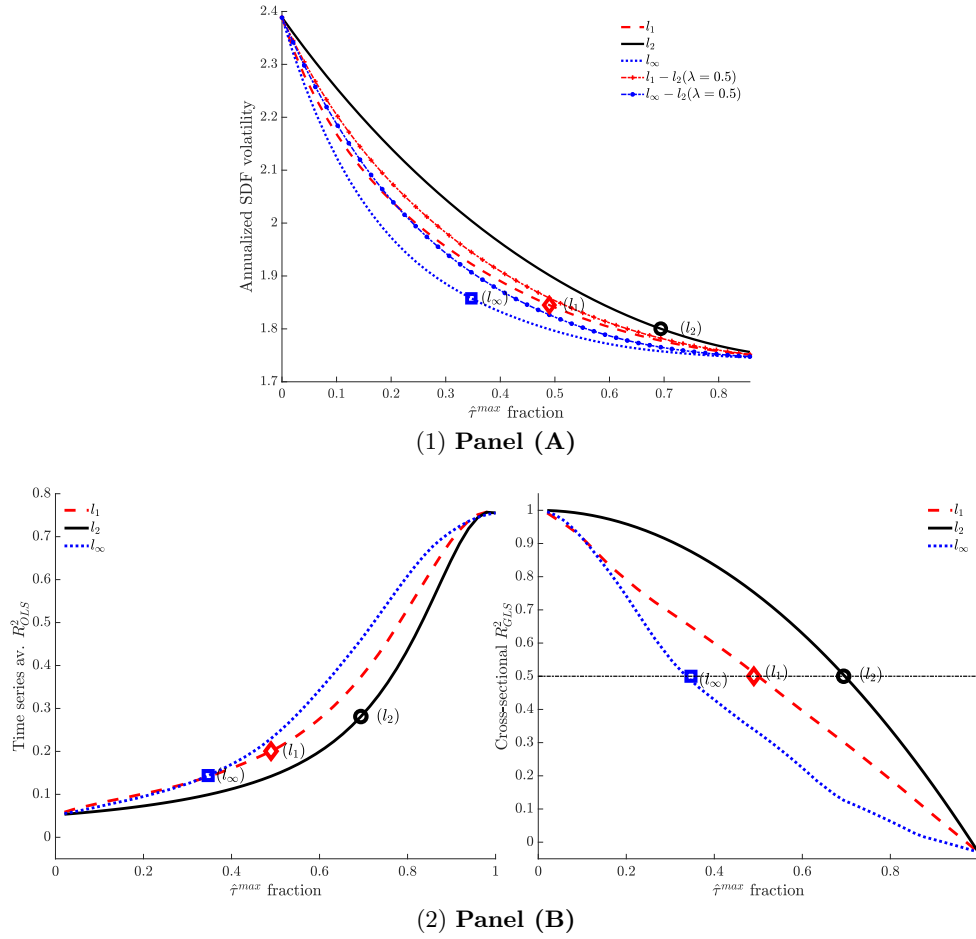


Figure 2: Minimum APT S-SDF volatility curve and tradeoff between time-series and cross-sectional S-SDF explanatory power. Panel (A) reports the annualized volatility of minimum variance APT S-SDFs as a function of the relative threshold $\kappa \in (\tau^{min}/\tau^{max}, 1]$, with threshold τ^{max} defined in equation (26). We consider the pricing metrics $h = \|\cdot\|_1$ (dashed line), $h = \|\cdot\|_2$ (solid line) and $h = \sqrt{N_D} \|\cdot\|_\infty$ (dotted line). The two additional curves report the minimum S-SDF volatility curve for pricing metrics $h = \lambda \|\cdot\|_1 + (1 - \lambda) \|\cdot\|_2$ (dashed-crossed line) and $h = \sqrt{N_D} \lambda \|\cdot\|_\infty + (1 - \lambda) \|\cdot\|_2$ (dashed dotted line), where $\lambda = 0.5$. Panel (B) reports the average time series R^2 metric (32) (left panel) and the cross-sectional GLS R^2 metric (31) (right panel) for the first three minimum variance S-SDFs from Panel (A). Finally, the diamonds, the circles, and the squares in Panel (B) identify the corresponding S-SDFs attaining a cross-sectional GLS R^2 of 50%. Sorted portfolio returns are used to construct the S-SDFs with traded risk factor given by the market return. All calculations are based on the low dimensional dataset from July 1963 to June 2018.

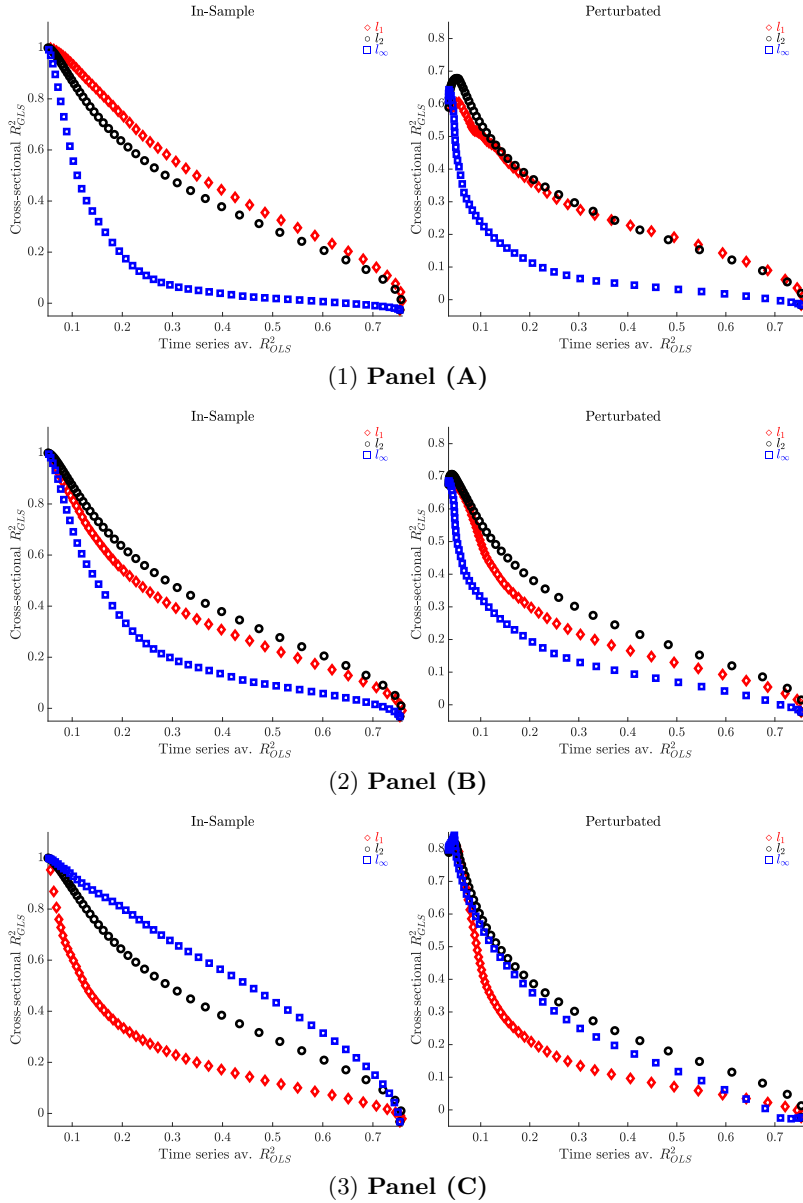


Figure 3: Tradeoff between time series and cross-sectional APT S-SDF explanatory power. The figure reports the average time-series R^2 metric (32) and cross-sectional GLS R^2 metric (33), using the l_1 -norm in **Panel (A)**, the l_2 -norm in **Panel (B)**, and the (scaled) l_∞ -norm in **Panel (C)**, for minimum variance APT S-SDFs based on a l_1 -, l_2 -, and (scaled) l_∞ -pricing metric, while taking the market excess return as the single traded risk factor. Left panels report in-sample results with S-SDFs estimated and evaluated on the same 40 year window 1965-2005, while right panels report results with S-SDFs estimated on 40 year window 1965-2005 and evaluated on 40 year window 1975-2015. All computations are based on the low dimensional dataset.

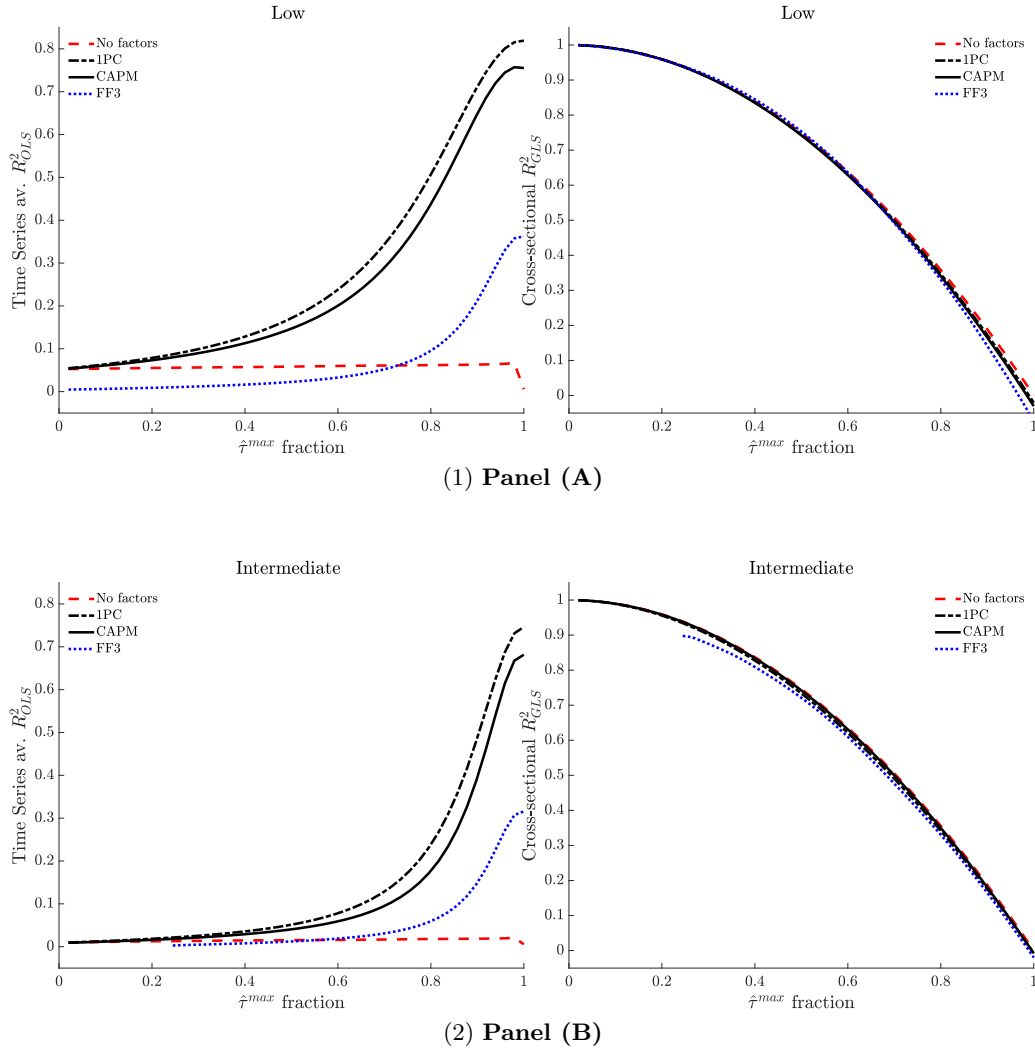


Figure 4: Role of traded factors in minimum dispersion APT S-SDFs. The figure reports the average time-series R^2 metric (32) (left panel) and cross-sectional GLS R^2 metric (31) for various minimum variance APT S-SDFs, corresponding to different choices for the set of traded risk factors, in the low dimensional (**Panel (A)**) and intermediate dimensional (**Panel (B)**) datasets. We consider following choices for the traded factors: (1) No factors; (2) CAPM, corresponding to the market return as the single traded factor; (3) FF3, corresponding to the three Fama-French factor returns and traded factors; (4) 1PC, corresponding to the first principal component of excess returns as the single traded factor. All calculations are for APT S-SDFs based on the standard l_2 -pricing metric. Both the low and intermediate dimensional datasets run from July 1963 to June 2018.

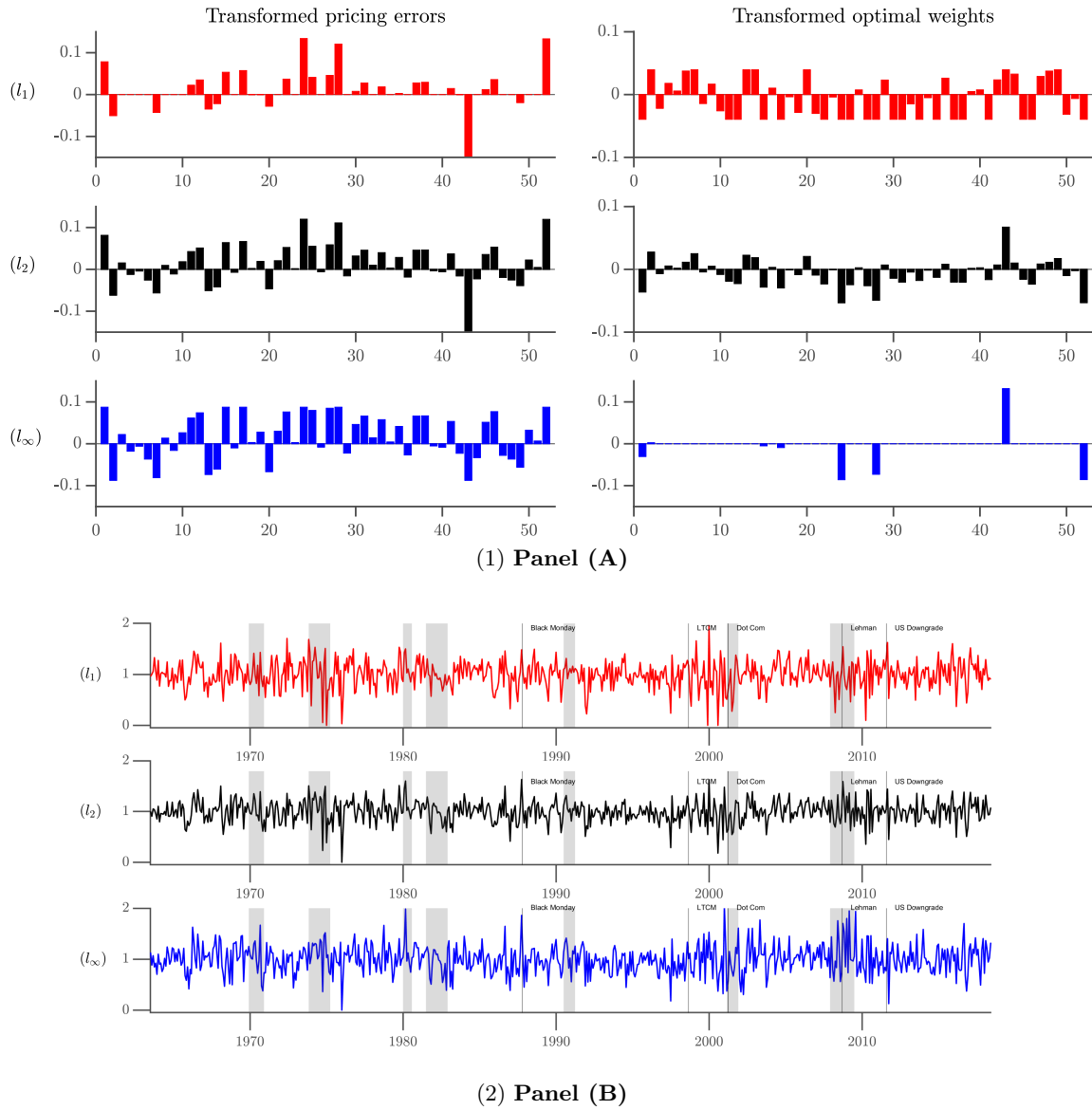


Figure 5: **Properties of APT S-SDFs.** Panel (A) reports, from top to bottom, $\Sigma_\zeta^{-1/2}$ -transformed pricing errors (left) and $\Sigma_\zeta^{1/2}$ -transformed optimal portfolio weights (right) of the three minimum variance APT S-SDFs highlighted with a diamond, a circle and a square in Panel (B) of Figure 2, which correspond to a l_1 - (first row), l_2 - (second row), and scaled l_∞ - (third row) APT pricing metric, respectively. Panel (B) reports the time-series of these three minimum variance APT S-SDFs. All calculations are based on the low dimensional dataset from July 1963 to June 2018, taking the market excess return as the single traded factor. Grey shaded areas in Panel (B) highlight NBER recession periods.

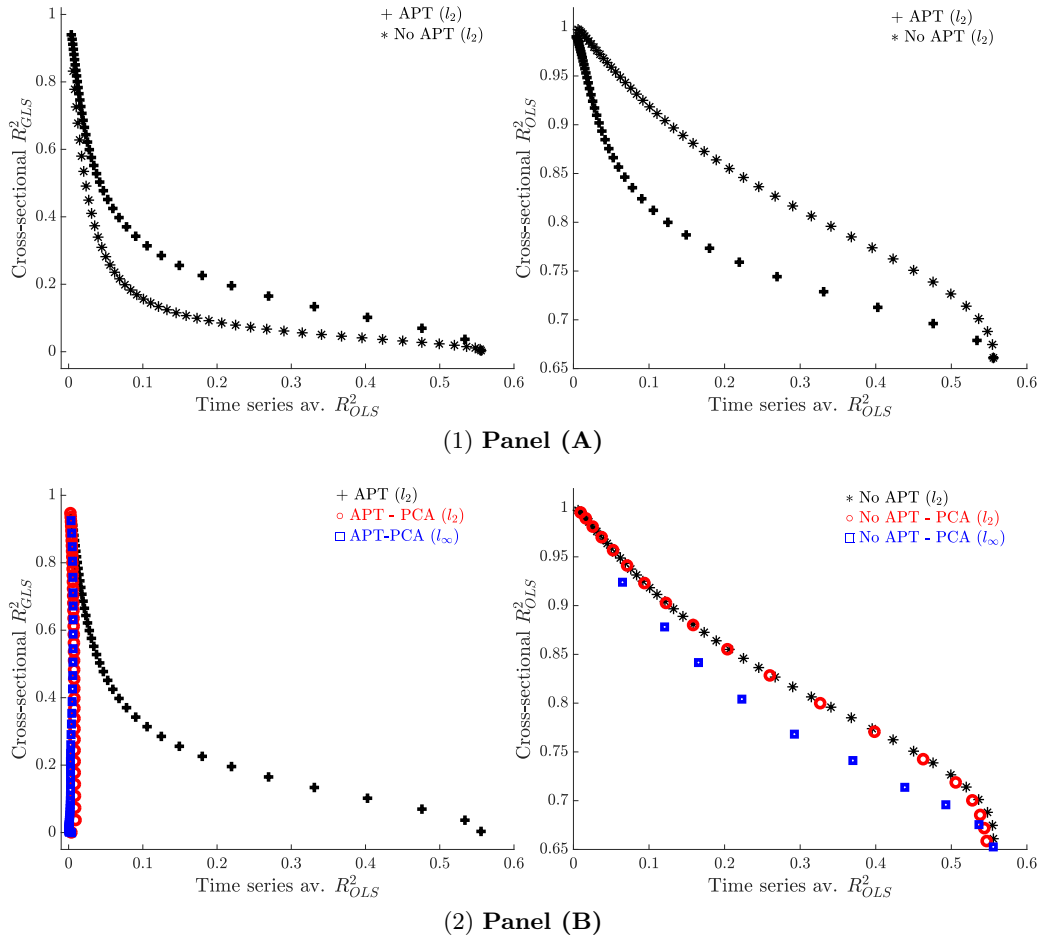


Figure 6: **Tradeoff between time-series and cross-sectional S-SDF explanatory power of APT S-SDFs, non APT S-SDFs, and principal components based S-SDFs.** For various minimum variance S-SDFs, the figure reports the tradeoff between average time series R^2 metric (32) and cross-sectional GLS R^2 metric (31) (denoted by R^2_{GLS} , left panels) or cross-sectional R^2 metric (34) (denoted by R^2_{OLS} , right panels). The following S-SDFs are compared in Panel (A). First, APT S-SDFs based on the l_2 -pricing metric. Second, non APT S-SDFs based on the l_2 -pricing metric, but with a diagonal weighting matrix $\Sigma_\eta = \mathbf{I}_{N_D \times N_D}$. Both these S-SDFs take the first principal component of excess returns as the single traded factor. In each figure of Panel (B), results for two additional principal components based S-SDFs are reported, which impose the corresponding pricing constraint on the principal components of sorted portfolio excess returns, instead of the original excess returns, without incorporating any traded factor. The first and second of these S-SDFs impose in the left (right) figure an APT l_2 - and l_∞ - pricing constraint (a non APT l_2 - and l_∞ - pricing constraint based on a diagonal weighting matrix $\Sigma_\eta = \mathbf{I}_{N_D \times N_D}$), respectively. All results are based on the high dimensional dataset running from 1970 to 2018.

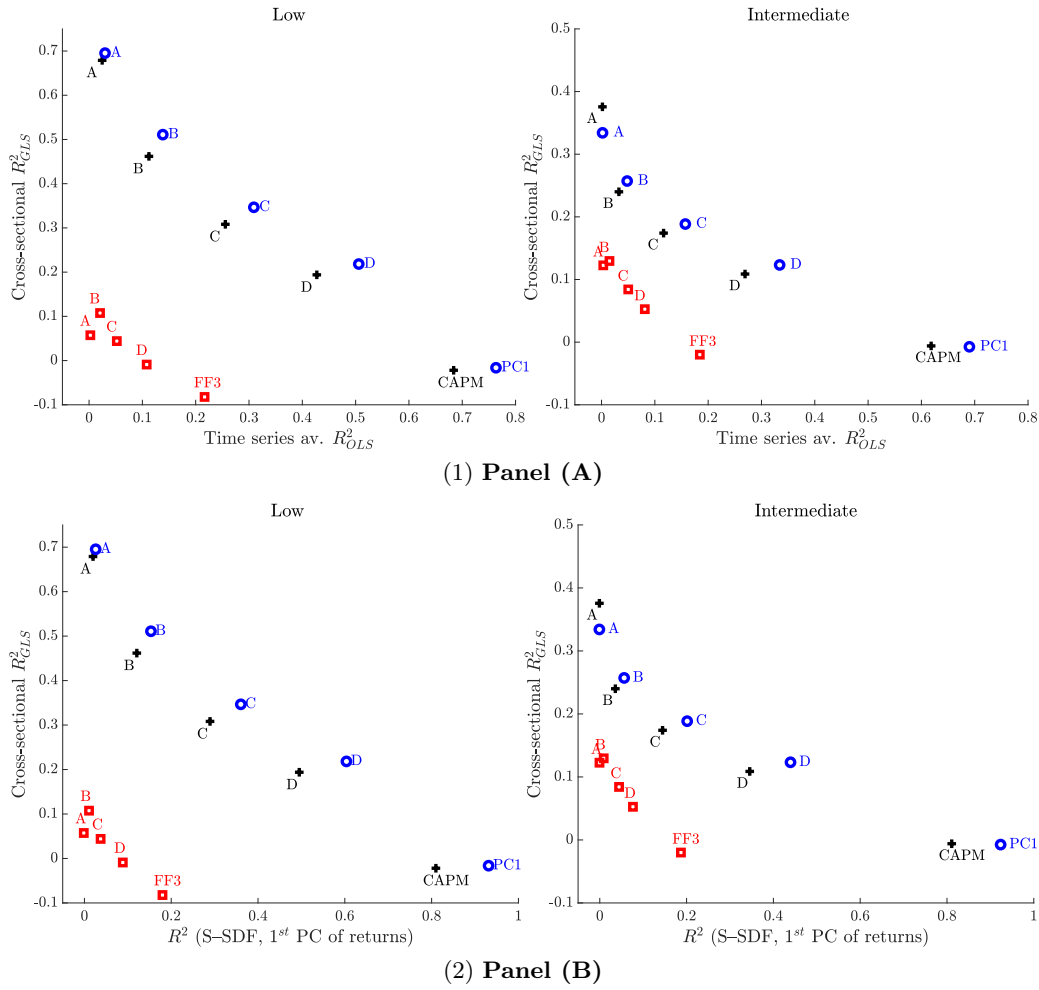


Figure 7: **Out-of-sample tradeoff between time-series and cross-sectional explanatory power of optimal data-driven APT S-SDFs.** The figure reports in **Panel (A)** (**Panel (B)**) out-of-sample average time-series R^2 metric (32) (out-of-sample time-series R^2 metric (32) with respect to the first principal component of excess returns) and out-of-sample cross-sectional GLS R^2 metric (31), for the optimal data-driven minimum variance APT S-SDFs with l_2 -pricing metric from Section 5.7. These S-SDFs are estimated on rolling windows of 30 years by maximizing the in-sample cross-sectional metric $R^2_{GLS}(\hat{\kappa}_y)$ in constrained optimization problem (35), for parameters $p = 0$ (label A), $p = 0.25$ (label B), $p = 0.5$ (label C), $p = 0.75$ (label D), and $p = 1$. APT S-SDFs are based on three choices for the traded factor returns: (1) the market excess return, (2) the three Fama-French factor excess returns, and (3) the first principal component of sorted portfolio excess returns. Label CAPM corresponds to case (1) for $p = 1$, label FF3 to case (2) for $p = 1$ and label PC1 to case (3) for $p = 1$. Results are based on the low dimensional (left panels) and the intermediate dimensional (right panels) datasets.

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Smart Stochastic Discount Factors

ONLINE APPENDIX

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In this Online Appendix, we collect the proofs of the theoretical statements in the main text, additional theoretical findings and useful auxiliary results to the theory in the main text, and further empirical evidence on S-SDFs complementing the one presented in the main text. Section A first reports proofs of propositions and corollaries in the main text. It then collects auxiliary results not fully discussed in the main text, together with the corresponding proofs. Section B summarizes the properties of ϕ -dispersions in the Cressie-Read family, details further areas of application of our S-SDF framework, and provides the foundations of S-SDFs in viable markets with convex cost function σ . Section C collects complementary empirical evidence on minimum dispersion APT S-SDFs. Section D collects figures illustrating the additional empirical evidence in Section C.

A ONLINE APPENDIX – Proofs

Proof of Proposition 1. By [Rockafellar, 1970, Thm. 13.2] σ_C given in equation (3) of the main text is proper, closed and sublinear.¹ This implies that the pricing functional π is sublinear and that the set of payoffs, $\mathcal{Z}$ is a convex cone. Hence, by [Chen, 2001, Thm. 1, 5 and Cor. 1], absence of free lunches is equivalent to the existence of an almost surely strictly positive SDF, M , such that $\mathbb{E}[MZ] \leq \pi(Z)$ for any $Z \in \mathcal{Z}$.²

By definition of cost functional π and payoff space $\mathcal{Z}$, it equivalently follows for any $\theta \in \mathbb{R}^N$ that $\theta'P + \sigma_C(\theta) \geq \mathbb{E}[M\theta'X]$, i.e., $\theta'(\mathbb{E}[MX] - P) \leq \sigma_C(\theta)$. By [Bauschke and Combettes, 2011, Prop. 13.10 (i)] this inequality holds if and only if

$$(\sigma_C)^*(\mathbb{E}[MX] - P) \leq 0 . \quad (\text{OA-1})$$

By [Rockafellar, 1970, Thm. 13.2], the convex conjugate σ_C^* is given by the character-

¹Properness follows since constrained set C is not empty.

²Technically, in order to use [Chen, 2001, Thm. 5], we require that there exists a traded payoff that is strictly positive almost surely. Such assumption is easily satisfied, e.g., when there exists a risk-free asset with positive risk-free payoff.

istic function δ_C , i.e.,

$$\sigma_C^*(\mathbb{E}[M\mathbf{X}] - \mathbf{P}) = \begin{cases} 0 & \mathbb{E}[M\mathbf{X}] - \mathbf{P} \in C \\ +\infty & \text{else} \end{cases}.$$

Thus, inequality (OA-1) is equivalent to condition $\mathbb{E}[M\mathbf{X}] - \mathbf{P} \in C$. $\square$

Proof of Corollary 1. By Proposition 1, absence of free lunches in market $(\tilde{Z}_{\{0\}}, \tilde{\pi}_{\{0\}}, \tilde{\mathbb{P}})$ is equivalent to the existence of a strictly positive SDF $\tilde{M}$ such that $\tilde{\mathbb{E}}[\tilde{M}\mathbf{X}] = \mathbf{P}$. Moreover, the absence of free lunches in market $(Z_C, \pi_C, \mathbb{P})$ is equivalent to the existence of a strictly positive SDF M such that $\mathbb{E}[M\mathbf{X}] - \mathbf{P} \in C$. $\square$

Proof of Proposition 2. We first prove that $\Pi(C) = -\Delta(C)$. To do so, we rewrite problem $\Pi(C)$ according to the notation given in Bauschke et al. [2017] and obtain its Fenchel-Rockafellar's dual problem. Let $A : L^q \rightarrow \mathbb{R}^N$ be the linear function $A(M) := \mathbb{E}[M\mathbf{X}]$, and let $g : \mathbb{R}^N \rightarrow (-\infty, +\infty]$ denote the function $g(\boldsymbol{\eta}) := \delta_R(\boldsymbol{\eta})$ with $R := \mathbf{P} + C = \{\mathbf{P} + \boldsymbol{\eta} : \boldsymbol{\eta} \in C\}$. With the notation $I_f := \mathbb{E}[f(\cdot)]$ for any function $f : \mathbb{R} \rightarrow [-\infty, +\infty]$, it follows that $I_{\phi_+}(M) = I_\phi(M) + \delta_{L^q_+}(M)$.³ Thus we can write:

$$\Pi(C) = \inf_{M \in L^q} \{I_{\phi_+}(M) + g(A(M))\}.$$

As payoffs are in L^p with $1/p + 1/q = 1$, A is continuous, while the properties of ϕ_+ imply that I_{ϕ_+} is a closed convex function. Since C is convex, closed and non-empty, g is convex, closed and proper. In view of these properties, we can obtain the dual problem of Π via [Bauschke et al., 2017, Thm. 15.23]. In order to apply the mentioned theorem, [Bauschke

³Remember that ϕ_+ is the restriction of ϕ to $\mathbb{R}_+$, i.e., $\phi_+(x) = \phi(x)$ if $x \geq 0$ and $\phi_+(x) = +\infty$ if $x < 0$.

et al., 2017, Prop. 15.24] shows that we only need to check that:⁴

$$\text{ri}(\text{dom}(g)) \cap A[\text{qri} \text{ dom}(I_{\phi_+})] \neq \emptyset . \quad (\text{OA-2})$$

As shown in [Borwein and Lewis, 1991, Cor. 2.6], our requirement $(0, +\infty) \subset \text{dom } \phi$ implies that $A(\text{dom } I_\phi \cap L_{++}^q) \subset A[\text{qri}(\text{dom } I_{\phi_+})]$. Hence, since $\text{dom}(g) = R$, showing that $A(\text{dom } I_\phi \cap L_{++}^q) \cap \text{ri}(R) \neq \emptyset$ is enough to prove (OA-2). Under the assumption that the economy $(\mathcal{Z}_{\tilde{C}}, \pi_{\tilde{C}}, \mathbb{P})$ admits no free lunches, with $\tilde{C} \subset \text{ri}(C)$, by Proposition 1 there exists $\tilde{M} \in \text{dom } I_\phi \cap L_{++}^q$, hence $A(\tilde{M}) \in A(\text{dom } I_\phi \cap L_{++}^q)$. Moreover, $\mathbb{E}[\tilde{M}\mathbf{X}] \in \mathbf{P} + \tilde{C}$, i.e., $A(\tilde{M}) \in \mathbf{P} + \tilde{C} \subset \text{ri}(R)$ holds and condition (OA-2) is satisfied.

Thus, by [Bauschke et al., 2017, Thm. 15.23 and 15.24] we obtain $\Pi(C) = -\Delta(C)$ where

$$\Delta(C) = - \min_{\boldsymbol{\theta} \in \mathbb{R}^N} \{I_{\phi_+}^*(-{}^t A(\boldsymbol{\theta})) + g^*(\boldsymbol{\theta})\} , \quad (\text{OA-3})$$

where $I_{\phi_+}^* : L^p \rightarrow [0, +\infty]$ is the conjugate function of I_{ϕ_+} and ${}^t A : \mathbb{R}^N \rightarrow L^p$ is the adjoint map of A , given by ${}^t A(\boldsymbol{\theta}) = \mathbf{X}'\boldsymbol{\theta}$.⁵ As ϕ_+ is convex and closed, we can apply [Rockafellar, 1968, Thm. 2] to obtain $I_{\phi_+}^* = I_{\phi_+}^*$. Moreover, for every $\boldsymbol{\theta} \in \mathbb{R}^N$,

$$\begin{aligned} g^*(-\boldsymbol{\theta}) &= \delta_R^*(-\boldsymbol{\theta}) \\ &= \sup_{\boldsymbol{\eta} \in \mathbb{R}^N} \{-\boldsymbol{\theta}'\boldsymbol{\eta} : \boldsymbol{\eta} \in \mathbf{P} + C\} \\ &= -\mathbf{P}'\boldsymbol{\theta} + \sigma_C(-\boldsymbol{\theta}) . \end{aligned}$$

Thus (OA-3) reads:

$$\Delta(C) = - \min_{\boldsymbol{\theta} \in \mathbb{R}^N} \{ \mathbb{E}[\phi_+^*(-\mathbf{X}'\boldsymbol{\theta})] + \mathbf{P}'\boldsymbol{\theta} + \sigma_C(\boldsymbol{\theta}) \} ,$$

⁴See [Bauschke et al., 2017, Def. 6.9] for the definition of relative interior, ri, and quasi relative interior, qri.

⁵The adjoint map of A , ${}^t A$, is characterized by the identity $\mathbb{E}[{}^t A(\boldsymbol{\theta})M] = \boldsymbol{\theta}'\mathbb{E}[M\mathbf{X}]$, for each $M \in L^q$ and each portfolio weights $\boldsymbol{\theta} \in \mathbb{R}^N$.

thereby proving strong duality between $\Pi(C)$ and $-\Delta(C)$.

We now prove the relation between the optimal solutions of $\Pi(C)$ and $\Delta(C)$ given in equation (12) of the main text. By [Bauschke et al., 2017, Prop. 19.4], if θ_0 is a dual solution such that $I_{\phi_+^*}$ is Gateaux differentiable in $-\mathbf{X}'\theta_0$, then the primal problem either has no solution or it has a unique solution given by $\nabla I_{\phi_+^*}(-\mathbf{X}'\theta_0)$. Hereafter we show that indeed $I_{\phi_+^*}$ is Gateaux differentiable and that the derivative is given by $M_0 = (\phi_+^*)'(-\mathbf{X}'\theta_0)$.

The assumption that $-\mathbf{X}'\theta_0 < \lim_{y \rightarrow \infty} \phi(y)/y$ a.s., by [Borwein and Lewis, 1991, Lem. 4.2], implies that $-\mathbf{X}'\theta_0$ belongs a.s. to the interior of the domain of ϕ_+^* . This and the strict convexity of ϕ_+ on its domain imply that ϕ_+^* is differentiable a.s. in $-\mathbf{X}'\theta_0$, see [Borwein and Lewis, 1991, Thm. 4.6]. Let us denote such derivative by $M_0(\omega) := (\phi_+^*)'(-\mathbf{X}'(\omega)\theta_0)$. Hereafter in order to prove that $I_{\phi_+^*}$ is Gateaux differentiable in $-\mathbf{X}'\theta_0$, we show that M_0 is the unique element of $\partial I_{\phi_+^*}(-\mathbf{X}'\theta_0)$. Now let us consider any element $\bar{M} \in \partial I_{\phi_+^*}(-\mathbf{X}'\theta_0)$, then by [Rockafellar, 1970, Thm. 23.5]:

$$I_{\phi_+^*}(-\mathbf{X}'\theta_0) + I_{\phi_+}(\bar{M}) = \langle -\mathbf{X}'\theta_0, \bar{M} \rangle .$$

Explicitly, this gives $\int [\phi_+^*(-\mathbf{X}'(\omega)\theta_0) + \phi_+(\bar{M}(\omega)) + \mathbf{X}'(\omega)\theta_0\bar{M}(\omega)] d\mathbb{P}(\omega) = 0$. Here, the integrand, since by Fenchel's inequality is nonnegative, is zero a.s.. Applying again [Rockafellar, 1970, Thm. 23.5] to this integrand, we obtain $\bar{M}(\omega) \in \partial\phi_+^*({}^tA(-\theta_0)(\omega))$ a.s.. However, since ϕ_+^* is differentiable a.s. in $-\mathbf{X}'\theta_0$, $\partial\phi_+^*({}^tA(-\theta_0)(\omega))$ has a unique element. Hence, we have that $\bar{M} = M_0$ a.s., which proves that $\partial I_{\phi_+^*}(-\mathbf{X}'\theta_0)$ has a unique element and that $I_{\phi_+^*}$ is Gateaux differentiable in $-\mathbf{X}'\theta_0$ with derivative given by $(\phi_+^*)'(-\mathbf{X}'\theta_0)$. This concludes the proof. $\square$

Proof of Corollary 2. Given constraint set $C = \{\mathbf{0}_{N_S+1}\} \times \{\boldsymbol{\eta} \in \mathbb{R}^{N_D} : \|\boldsymbol{\eta}\|_{2,\Sigma_\zeta^{-1/2}} \leq \tau\}$, the associated penalization function is given by

$$\sigma_C(\boldsymbol{\theta}) = \sup_{\boldsymbol{\eta} \in C} \boldsymbol{\eta}'\boldsymbol{\theta} = \tau \|\boldsymbol{\theta}_D\|_{2,\Sigma_\zeta^{1/2}}$$

using the definition of dual norm and the self-conjugation of the l_2 -norm. Clearly, $\text{dom}(\sigma_C) = \mathbb{R}^N$, hence by [Clark, 1993, Thm. 6], the no free lunches condition in market $(\mathcal{Z}_C, \pi_C, \mathbb{P})$ is equivalent to the standard no arbitrage condition. Moreover, by Proposition 1, absence of free lunches in this market is equivalent to the existence of strictly positive S-SDF, M_{APT} , such that $\mathbb{E}[M_{APT}] = 1$, $\mathbb{E}[M_{APT}\mathbf{F}^e] = \mathbf{0}$ and $\|\mathbb{E}[M_{APT}\mathbf{R}^e]\|_{2, \Sigma_\zeta^{-1/2}} \leq \tau$. $\square$

Proof of Corollary 3. The relative interior of $C = \{\mathbf{0}_{N_S+1}\} \times \{\boldsymbol{\eta} \in \mathbb{R}^{N_D} : \|\boldsymbol{\eta}\|_{\Sigma_\zeta^{-1/2}} \leq \tau\}$ contains $\{\mathbf{0}_{N_S+1}\} \times \{\boldsymbol{\eta} \in \mathbb{R}^{N_D} : \|\boldsymbol{\eta}\|_{2, \Sigma_\zeta^{-1/2}} < \tau\}$, which in turn contains $\tilde{C}$. Since $\tilde{C}$ is convex, closed and contains $\mathbf{0}$, the result follows from Proposition 2, where in dual portfolio problem $\Delta(C)$ the penalization function is given by

$$\sigma_C(\theta) = \sup_{\boldsymbol{\eta} \in C} \boldsymbol{\eta}'\theta = \tau \|\theta_D\|_{*, \Sigma_\zeta^{1/2}}.$$

$\square$

Proposition OA-1 (Equivalence with dispersion constrained minimum pricing error problems). Let $C_\tau := \{\boldsymbol{\eta} \in \mathbb{R}^N : h(\boldsymbol{\eta}) \leq \tau\}$ where $\tau \in \mathbb{R}$ and h is a convex, proper and closed function. Suppose that $\Pi(C_\tau)$ and

$$\mathcal{P}(\nu) := \inf_{M \in \mathcal{M}_+} \{h(\mathbb{E}[M\mathbf{X} - \mathbf{P}]) : \mathbb{E}[\phi(M)] \leq \nu\} \quad (\text{OA-4})$$

with $\nu \geq 0$ are finite and attained. (i) If M_0 solves $\Pi(C_\tau)$, then there exists $\nu_0 \geq 0$ such that M_0 solves $\mathcal{P}(\nu_0)$. (ii) If M_0 is the unique solution of $\mathcal{P}(\nu)$, then there exists $\tau_0 \in \mathbb{R}$ such that M_0 solves $\Pi(C_{\tau_0})$.

Proof. (i) Let $\nu_0 := \mathbb{E}[\phi(M_0)]$. By strict convexity of ϕ and [Borwein and Lewis, 1991, Prop. 2.11], M_0 is the unique solution of $\Pi(C_\tau)$. Therefore, M_0 is the unique element of $\{M \in \mathcal{M}_+ : \mathbb{E}[\phi(M)] \leq \nu_0, h(\mathbb{E}[M\mathbf{X} - \mathbf{P}]) \leq \tau\}$. Thus, M_0 solves $\mathcal{P}(\nu_0)$. (ii) Let $\tau_0 := h(\mathbb{E}[M_0\mathbf{X} - \mathbf{P}])$. If M_0 is the unique solution of $\mathcal{P}(\nu)$, it is the unique element of $\{M \in \mathcal{M}_+ : \mathbb{E}[\phi(M)] \leq \nu, h(\mathbb{E}[M\mathbf{X} - \mathbf{P}]) \leq \tau_0\}$. Thus, M_0 solves $\Pi(C_{\tau_0})$. $\square$

Lemma OA-1. Consider norms h_1 and h_∞ in equations (27) and (28) of the main text. Then, penalization $\sigma_{C_i}(\boldsymbol{\theta}) = \tau h_{i*, \Sigma_\zeta^{1/2}}(\boldsymbol{\theta})$ ($i = 1, \infty$) from Corollary 3 in the main text is given in closed-form as follows.

(i) If $\lambda = 1$: $h_{1*}(\boldsymbol{\theta}) = h_{\infty*}(\boldsymbol{\theta}) = \|\boldsymbol{\theta}\|_2$.

(ii) If $\lambda = 0$: $h_{1*}(\boldsymbol{\theta}) = \|\boldsymbol{\theta}\|_\infty$ and $h_{\infty*}(\boldsymbol{\theta}) = \|\boldsymbol{\theta}\|_1 / \sqrt{N}$.

(iii) If $\lambda \in (0, 1)$:

$$h_{1*}(\boldsymbol{\theta}) = \min_{\mathbf{z} \in \mathbb{R}^N} \left\{ \max \left\{ \frac{\|\mathbf{z}\|_\infty}{1-\lambda}, \frac{\|\boldsymbol{\theta} - \mathbf{z}\|_2}{\lambda} \right\} \right\},$$

$$h_{\infty*}(\boldsymbol{\theta}) = \min_{\mathbf{z} \in \mathbb{R}^N} \left\{ \max \left\{ \frac{\|\mathbf{z}\|_1}{\sqrt{N}(1-\lambda)}, \frac{\|\boldsymbol{\theta} - \mathbf{z}\|_2}{\lambda} \right\} \right\}.$$

Proof of Lemma OA-1. Since the cases $\lambda = 0, 1$ are obvious, let $\lambda \in (0, 1)$. Denoting by $\|\cdot\|_A, \|\cdot\|_B$ two norms, following identity holds for any $\boldsymbol{\eta} \in \mathbb{R}^N$:

$$\|\boldsymbol{\eta}\| := \lambda \|\boldsymbol{\eta}\|_A + (1-\lambda) \|\boldsymbol{\eta}\|_B = \left\| \begin{pmatrix} \lambda \|\boldsymbol{\eta}\|_A \\ (1-\lambda) \|\boldsymbol{\eta}\|_B \end{pmatrix} \right\|_1.$$

From [Combettes et al., 2019, Thm. 2.5], we obtain for any $\boldsymbol{\theta} \in \mathbb{R}^N$:

$$\begin{aligned} \|\boldsymbol{\theta}\|_* &= \inf_{(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2) \in \mathbb{R}^{2N}} \left\{ \left\| \begin{pmatrix} \frac{1}{\lambda} \|\boldsymbol{\theta}_1\|_{A*} \\ \frac{1}{1-\lambda} \|\boldsymbol{\theta}_2\|_{B*} \end{pmatrix} \right\|_{1*} : \boldsymbol{\theta}_1 + \boldsymbol{\theta}_2 = \boldsymbol{\theta} \right\} \\ &= \min_{\mathbf{z} \in \mathbb{R}^N} \left\{ \max \left\{ \frac{1}{\lambda} \|\mathbf{z}\|_{A*}, \frac{1}{1-\lambda} \|\boldsymbol{\theta} - \mathbf{z}\|_{B*} \right\} \right\}, \end{aligned}$$

since $\|\cdot\|_{1*} = \|\cdot\|_\infty$. The statement of the lemma finally follows by further recalling that $\|\cdot\|_{2*} = \|\cdot\|_2$ and $\|\cdot\|_{\infty*} = \|\cdot\|_1$. $\square$

B ONLINE APPENDIX – Theory of S–SDFs and Applications

B.1 Cressie-Read Dispersions

We collect for easy reference relevant explicit examples of ϕ –dispersions from the Cressie-Read family, together with their corresponding functions ϕ_+ and ϕ_+^* ; see also [Kitamura et al. \[2004\]](#) and [Newey and Smith \[2004\]](#), among others.

Examples 1. 1. Kullback-Leibler dispersion:

$$\phi_+(x) = \begin{cases} x \ln x - x + 1 & x > 0 \\ 1 & x = 0 \\ +\infty & x < 0 \end{cases}, \quad \text{with } \phi_+^*(y) = \exp(y) - 1.$$

2. Negative entropy:

$$\phi_+(x) = \begin{cases} -\ln x + x - 1 & x > 0 \\ +\infty & x \leq 0 \end{cases}, \quad \text{with } \phi_+^*(y) = \begin{cases} -\ln(1 - y), & y < 1 \\ +\infty & y \geq 1 \end{cases}.$$

3. Power dispersion:

(a) for $\gamma > 1$ and $\beta = \gamma/(\gamma - 1)$,

$$\phi_+(x) = \begin{cases} x^\gamma/\gamma & x \geq 0 \\ +\infty & x < 0 \end{cases}, \quad \text{with } \phi_+^*(y) = \max\{y, 0\}^\beta/\beta,$$

(b) for $0 < \gamma < 1$ and $\beta = \gamma/(\gamma - 1)$,

$$\phi_+(x) = \begin{cases} \frac{\gamma x - x^\gamma}{1-\gamma} & x \geq 0 \\ +\infty & x < 0 \end{cases}, \quad \text{with} \quad \phi_+^*(y) = \begin{cases} (1 + y/\beta)^\beta - 1 & y < -\beta \\ +\infty & y \geq -\beta \end{cases},$$

(c) for $\gamma < 0$ and $\beta = \gamma/(\gamma - 1)$,

$$\phi_+(x) = \begin{cases} \frac{1-x^\gamma+\gamma(x-1)}{\gamma} & x > 0 \\ +\infty & x \leq 0 \end{cases}, \quad \text{with} \quad \phi_+^*(y) = \begin{cases} -\frac{(1-y)^\beta+1}{\beta} & y < 1 \\ +\infty & y \geq 1 \end{cases}.$$

In the above settings, function ϕ_+ is always closed and strictly convex, while function ϕ_+^* is always differentiable in the interior of its domain. Moreover, function ϕ_+^* is strictly convex in all cases, but case (3a).

The next lemma summarizes the analytical expressions for minimum dispersion S-SDFs corresponding to the particular choices of ϕ -dispersions from Examples 1.

Lemma OA-2. *Under the conditions of Proposition 2 in the main text, consider the Φ -dispersions induced by functions ϕ_+ in Examples 1. It then follows:*

$$(1) \ M_0 = \exp(-\mathbf{X}'\boldsymbol{\theta}_0).$$

$$(2) \ M_0 = 1/(1 + \mathbf{X}'\boldsymbol{\theta}_0), \text{ if } -\mathbf{X}'\boldsymbol{\theta}_0 < 1.$$

$$(3a) \ M_0 = \max\{-\mathbf{X}'\boldsymbol{\theta}_0, 0\}^{\beta-1}.$$

$$(3b) \ M_0 = (1 - \mathbf{X}'\boldsymbol{\theta}_0/\beta)^{\beta-1}, \text{ if } -\mathbf{X}'\boldsymbol{\theta}_0 < -\beta.$$

$$(3c) \ M_0 = (1 + \mathbf{X}'\boldsymbol{\theta}_0)^{\beta-1}, \text{ if } -\mathbf{X}'\boldsymbol{\theta}_0 < 1.$$

Proof of Lemma OA-2. For every ϕ -dispersion considered under Examples 1, ϕ_+ is strictly convex on its domain. Moreover, for ϕ -dispersions (1) and (3a) the interior of the domain of ϕ_+^* , $\text{int}(\text{dom } \phi_+^*)$ is the entire real line, i.e., there is no restriction on the

values of the optimal portfolio payoff. With regard to the other cases, the optimal portfolio payoff is restricted to be almost surely in the interior of the domain of ϕ_+^* , which is $(-\infty, 1)$ in cases (2) and (3c) and $(-\infty, -\beta)$ in case (3b), respectively. $\square$

B.2 S-SDFs induced by short selling constraints and bid-ask spreads

Minimum variance SDFs incorporating bid-ask spreads and short-sale constraints in arbitrage-free markets have been studied in Luttmer [1996]. In general, such market frictions can be modelled with a portfolio constraint $\theta \in K$, using a finitely generated convex cone $K \subset \mathbb{R}^N$. Luttmer [1996] shows that SDFs in such markets imply pricing errors constrained to belong to the polar cone of K , which we denote by K^0 .⁶ For instance, short-sale constraints are defined by non-negative portfolio positions, i.e., $K = \mathbb{R}_+^N$. Since the polar cone of $\mathbb{R}_+^N$ is $\mathbb{R}_-^N$, it follows that $\mathbb{E}[MX] \leq P$. In our framework, these market frictions are easily modelled via the characteristic function of K :

$$\sigma_{K^0}(\theta) = \delta_K(\theta) := \begin{cases} 0 & \theta \in K \\ +\infty & \theta \notin K \end{cases}.$$

Literally, investors incur in zero transaction costs if portfolio positions belong to K and arbitrarily large transaction costs otherwise. The S-SDF valuations obtained in Luttmer [1996] are constrained to belong to set $P + K^0$.

B.3 Theories of Good-Deal-Bounds and robust identification of investor beliefs

Consider first the Good-Deal Bounds theory in Cochrane and Saa-Requejo [2000], which specifies a maximal price $\bar{P}_D$ and a minimal price $\underline{P}_D$ for a newly issued security with payoff X_D , given an exact pricing condition for a set of traded securities and under an upper bound on the maximum attainable Sharpe Ratio. This bound is introduced to rule

⁶The polar cone of set K is the set $K^0 := \{y \in \mathbb{R}^N : x'y \leq 0, \text{ for all } x \in K\}$.

out unreasonably good deals in the extended market including the new security. Let M_S be the minimum dispersion SDF that minimizes the SDF dispersion for a given convex dispersion function ϕ and which prices exactly the set of sure securities.⁷ Therefore, M_S solves the optimization problem:

$$\inf\{\mathbb{E}[\phi(M)] : M \in \mathcal{M}_+\} .$$

Let $\nu_S := \mathbb{E}[\phi(M_S)]$ and denote by $P_D := \mathbb{E}[M_S X_D]$ the reference price for the newly issued payoff under SDF M_S . The minimal and maximal price for the newly issued security are obtained for an upper dispersion threshold $\nu \geq \nu_S$ as follows:

$$\underline{P}_D := P_D + \inf_{M \in \mathcal{M}_+} \{\mathbb{E}[M X_D] - P_D : \mathbb{E}[\phi(M)] \leq \nu\} , \quad (\text{OA-5})$$

$$\overline{P}_D := P_D - \inf_{M \in \mathcal{M}_+} \{-(\mathbb{E}[M X_D] - P_D) : \mathbb{E}[\phi(M)] \leq \nu\} . \quad (\text{OA-6})$$

By construction, these prices satisfy the inequalities $\underline{P}_D \leq P_D \leq \overline{P}_D$. Moreover, they are supported by corresponding S-SDFs $\underline{M}_0$ and $\overline{M}_0$, which solve each one of the minimizations on the RHS of identities (OA-5)-(OA-6):

$$\underline{M}_0 = \arg \min_{M \in \mathcal{M}_+} \{\underline{h}(\mathbb{E}[M X_D] - P_D) : \mathbb{E}[\phi(M)] \leq \nu\} , \quad (\text{OA-7})$$

$$\overline{M}_0 := \arg \min_{M \in \mathcal{M}_+} \{\overline{h}(\mathbb{E}[M X_D] - P_D) : \mathbb{E}[\phi(M)] \leq \nu\} , \quad (\text{OA-8})$$

with corresponding pricing metrics given by $\underline{h}(\boldsymbol{\eta}) = \boldsymbol{\eta}$ and $\overline{h} = -\boldsymbol{\eta}$. It thus follows from Proposition OA-1 of Appendix A that $\underline{P}_D$ and $\overline{P}_D$ can be equivalently obtained by computing the price of the new security's payoff X_D under these two minimum dispersion S-SDFs.

Consider now [Chen et al. \[2020\]](#) theory for a robust identification of investor beliefs.

⁷In [Cochrane and Saa-Requejo \[2000\]](#) Good-Deal Bounds theory the notion of dispersion minimized is the SDF variance.

This theory relies on the determination of a S–SDF $\underline{M}_0$ solving problem (OA-7) under robust notions of S–SDF dispersion (cases 1 and 3a of Section B.1) and while imposing the normalization constraint $\mathbb{E}[M] = 1$, which can be naturally incorporated in constrained set $\mathcal{M}_+$ as a sure pricing constraint. Such normalization gives rise to the interpretation of $\underline{M}_0$ as a density reproducing a recovered robust investor belief with respect to the reference probability used to describe risky asset payoffs. Such interpretation is further motivated by the fact that in this theory ”sure payoffs” $\mathbf{X}_S$ are not asset payoffs, but instead asset payoffs that have been already stochastically discounted by a parametric SDF M^* implied by economic theory. Therefore, in this setting $\mathbb{E}[\underline{M}_0 X_D]$ has the interpretation of a conservative expected payoff under the recovered investor belief, instead of a lowest arbitrage-free price for payoff X_D .

B.4 Existence of S–SDFs for markets with convex costs

We have shown in Section 2.2 of the main text that S–SDFs are characterized by the absence of free lunches in an economy with sublinear costs. In this section, we show that S–SDFs arise more generally also in an economy with convex costs, under a no-arbitrage condition stronger than the absence of free lunches, i.e., market viability, as defined in, e.g., Harrison and Kreps [1979].⁸ To this end, let portfolio positions involve costs measured by closed convex function $\sigma : \mathbb{R}^N \rightarrow [0, \infty]$. As in the main text, we define the set of traded payoffs as the set of portfolio payoffs involving finite costs:

$$\mathcal{Z} := \{Z = \mathbf{X}'\boldsymbol{\theta} : \sigma(\boldsymbol{\theta}) < +\infty\} .$$

Accordingly, we define a pricing functional π on $\mathcal{Z}$ by:

$$\pi(Z) := \inf_{\boldsymbol{\theta} \in \mathbb{R}^N} \{\mathbf{P}'\boldsymbol{\theta} + \sigma(\boldsymbol{\theta}) : Z = \mathbf{X}'\boldsymbol{\theta}\} .$$

⁸A market $(\mathcal{Z}, \pi, \mathbb{P})$ is viable if there exists an agent (represented by preference $\succsim$) and $Z^* \in \mathcal{Z}$ such that $\pi(Z^*) \leq 0$ and $Z^* \succsim Z$ for all $Z \in \mathcal{Z}$ with $\pi(Z) \leq 0$. The preference relation $\succsim$ is assumed to be convex, continuous and strictly increasing.

Since cost function σ is closed and convex, π is a convex pricing functional and $\mathcal{Z}$ is a closed and convex set of traded payoffs. With these definitions, the next proposition provides the closed-form supporting market in which the existence of a S-SDF is characterized by market viability.

Proposition OA-2. *Market $(\mathcal{Z}, \pi, \mathbb{P})$ is viable if and only if there exists an almost surely strictly positive SDF $M \in L^q(\mathbb{P})$, such that*

$$\mathbb{E}[MX] - \mathbf{P} \in C_\sigma ,$$

where

$$C_\sigma := \{\boldsymbol{\eta} \in \mathbb{R}^N : \boldsymbol{\eta}'\boldsymbol{\theta} \leq \sigma(\boldsymbol{\theta}) \text{ for all } \boldsymbol{\theta} \in \mathbb{R}^N\} .$$

Proof of Proposition OA-2. Convexity of σ implies convexity of pricing functional π and the set of traded payoffs $\mathcal{Z}$. By [Chen, 2001, Thm. 1 and Cor. 1], market viability is equivalent to the existence of an a.s. strictly positive SDF, M , such that $\mathbb{E}[MZ] \leq \pi(Z)$ for any $Z \in \mathcal{Z}$. Now, by definition of pricing functional π and traded payoff space $\mathcal{Z}$, it equivalently follows for any $\boldsymbol{\theta} \in \mathbb{R}^N$ that $\boldsymbol{\theta}'\mathbf{P} + \sigma(\boldsymbol{\theta}) \geq \mathbb{E}[M\boldsymbol{\theta}'\mathbf{X}]$, i.e.,

$$\boldsymbol{\theta}'(\mathbb{E}[MX] - \mathbf{P}) \leq \sigma(\boldsymbol{\theta}) . \tag{OA-9}$$

Equivalently, by definition of C_σ , we have $\mathbb{E}[MX] - \mathbf{P} \in C_\sigma$.

□

Given a closed convex set $C = \{\boldsymbol{\eta} \in \mathbb{R}^N : h(\boldsymbol{\eta}) \leq \tau\}$ defining a S-SDF with closed convex function h and threshold $\tau \geq 0$, the convex arbitrage-free economy supporting such constraint set C is based on a cost function given by:

$$\sigma = h^* + \tau .$$

While in Proposition 1 of the main text the necessary and sufficient condition for the existence of a strictly positive S–SDF in an economy with sublinear cost function σ_C is the absence of free-lunches, when h^* is not sublinear the no-arbitrage condition equivalent to the existence of a strictly positive S–SDF is market viability.

C ONLINE APPENDIX – Empirics of APT S–SDFs

C.1 S–SDF tradeoff between pricing accuracy and time series explanatory power

Figure OA-4 in Section D of this Online Appendix documents the effect of the choice of the notion of S–SDF dispersion for the S–SDF tradeoff between time series and cross-sectional explanatory powers. The two left panels of Figure OA-4 show that the resulting tradeoff of S–SDFs minimizing notions of dispersion such as Kullback-Leibler or Hellinger divergence is slightly Pareto dominated by minimum variance S–SDFs on non perturbed data.⁹ Moreover, minimum variance S–SDFs produce a clearly more consistent tradeoff on perturbed data, both for the low- and intermediate dimensional datasets.

C.2 Basic out-of-sample framework and evidence

For every in-sample window of 30 years of monthly observations with last observation in month y , the in-sample S–SDF estimation requires the specification of a corresponding threshold, denoted by $\hat{\tau}_y$.¹⁰ Similar to the in-sample analysis, our theoretical findings offer a natural way to determine the range of relevant empirical thresholds. We obtain the maximal admissible threshold $\hat{\tau}_y^{max}$ as the empirical version of equation (26) in the main

⁹Kullback-Leibler and Hellinger divergence are members of the family of Cressie-Read notions of stochastic dispersion detailed in Section B.1 of this Online Appendix (see case 1 and case 3b for $\gamma = 1/2$, respectively).

¹⁰Given our data sample, this gives us an out-of-sample period of monthly observations from July 1963 to June 2018. The first in-sample window of 30 years consists of monthly observations from July 1933 to June 1963.

text. For instance, under a set of traded risk factors including the market excess return alone, this corresponds to the empirical threshold implied by a minimum variance SDF that is only required to price exactly the market excess return in the given estimation window. The minimal admissible threshold $\hat{\tau}_y^{min}$ is obtained as in Section 5.3 in the main text, as the smallest threshold value for which in the given estimation window no empirical duality failure arises. This yields for any estimation window a range $[\hat{\tau}_y^{min}, \hat{\tau}_y^{max}]$ of admissible thresholds $\hat{\tau}_y$ and a range $[\hat{\kappa}_y^{min}, 1]$ of admissible relative thresholds $\hat{\kappa}_y$, where $\hat{\kappa}_y^{min} := \hat{\tau}_y^{min} / \hat{\tau}_y^{max}$. Given a sequence of admissible relative thresholds $\{\hat{\kappa}_y\}$, we apply the estimation methodology in Section 5.2 of the main text, to obtain a sequence $\{\hat{\theta}_y\}$ of estimated S-SDF dual portfolio weights, which is updated at a semi-annual frequency. For each of the six months following month y , we denote by $\mathbf{X}_{y+m}$ the vector of excess returns in month $y+m$ and estimate with Corollary 3 the corresponding sequence of out-of-sample monthly minimum variance S-SDFs, i.e., $\hat{M}_{y+m} := \max\{-\hat{\theta}_y' \mathbf{X}_{y+m}, 0\}$. In this way, we obtain a monthly time series $\{\hat{M}_{y+m}\}$ of data driven APT S-SDFs, for which we can evaluate the out-of-sample time series and cross-sectional explanatory power for the sequence of out-of-sample return vectors $\{\mathbf{X}_{y+m}\}$.

Regarding the choice of the sequence of relative thresholds $\{\hat{\kappa}_y\}$, we adopt in this section a simple approach based on constant relative thresholds $\hat{\kappa}_y = \bar{\kappa}$. This approach produces a useful description of the attainable out-of-sample tradeoffs between time series and cross-sectional S-SDF explanatory power, which may arise from data-driven threshold selections holding the relative threshold roughly constant across estimation windows, but it does not produce a criterion for a non forward-looking choice of an optimal threshold. The first two rows of Figure OA-5 report the resulting curves of out-of-sample average time series R^2 (31) and cross-sectional GLS R^2 (32), in dependence of the value of constant relative threshold $\bar{\kappa} \in [\max_y \{\hat{\kappa}_y^{min}\}, 1]$, for minimum variance S-SDFs incorporating the market excess return as the single traded risk factor. All out-of-sample GLS R^2 curves are inversely U-shaped. In the low (intermediate) dimensional data set, the maximal GLR R^2 is as high

as about 70% (45%) for the APT S–SDF with l_2 –pricing constraints, when the relative threshold $\bar{\kappa}$ is about 40% (65%). These are large increases in pricing accuracy, compared, e.g., to the mildly negative out-of-sample GLR R^2 s of an empirical CAPM S–SDF exactly pricing only the market return in-sample ($\bar{\kappa} = 1$).

Importantly, we observe a tradeoff between time series and cross-sectional S–SDF out-of-sample explanatory power, which is summarized in the two bottom panels of in Figure OA-5, for relative thresholds between the threshold value at which the maximum of the GLS R^2 curve is attained and the maximum admissible value $\bar{\kappa} = 1$. Similar to the in-sample findings of Section 5.4.2 in the main text, APT S–SDFs with pricing metric different from the l_2 –norm imply a Pareto dominated out-of-sample tradeoff between cross-sectional and time series explanatory power, further confirming that S–SDF sparsity is quite costly also in an out-of-sample perspective.

D ONLINE APPENDIX – Additional Figures

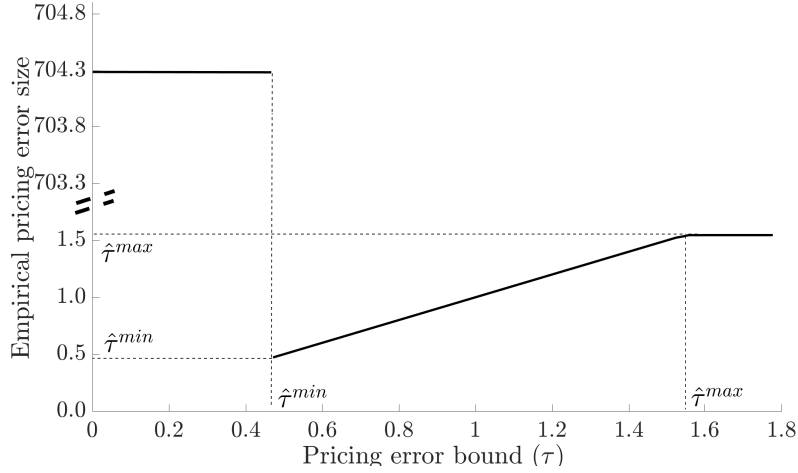


Figure OA-1: **Empirical duality failure.** We compute the minimum variance APT S–SDF M_0 with l_2 –pricing function for varying thresholds $\tau \geq 0$. On the y-axis we report the estimate of the function value $h(\mathbb{E}[M_0 \mathbf{X} - \mathbf{P}])$ for each τ . The point of discontinuity in the plot identifies the smallest threshold τ , for which a solution of the empirical primal S–SDF problem exists. The largest threshold $\hat{\tau}^{max}$ in the plot is computed as the sample version of the maximal threshold τ^{max} in equation (26) of the main text. Sorted portfolio returns are used to construct the S–SDFs without assuming any observable traded factor. All calculations are based on the intermediate dimensional dataset from June 1990 to June 2018.

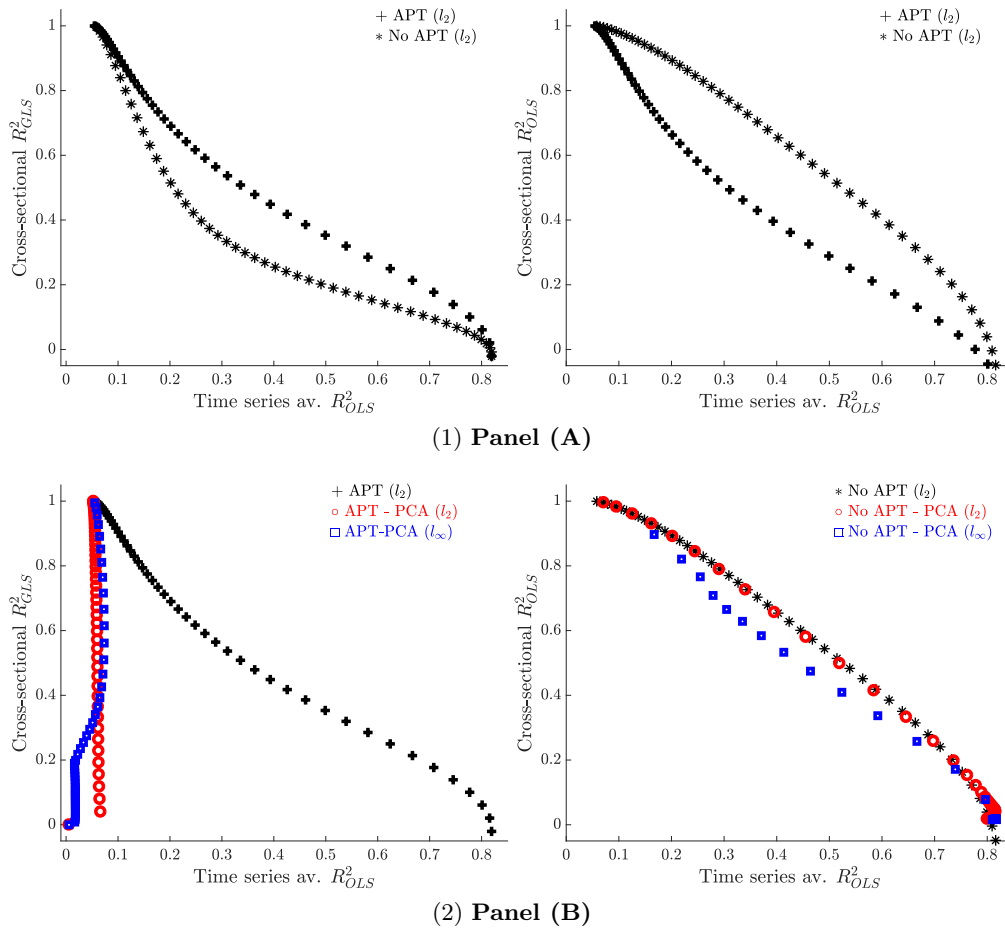
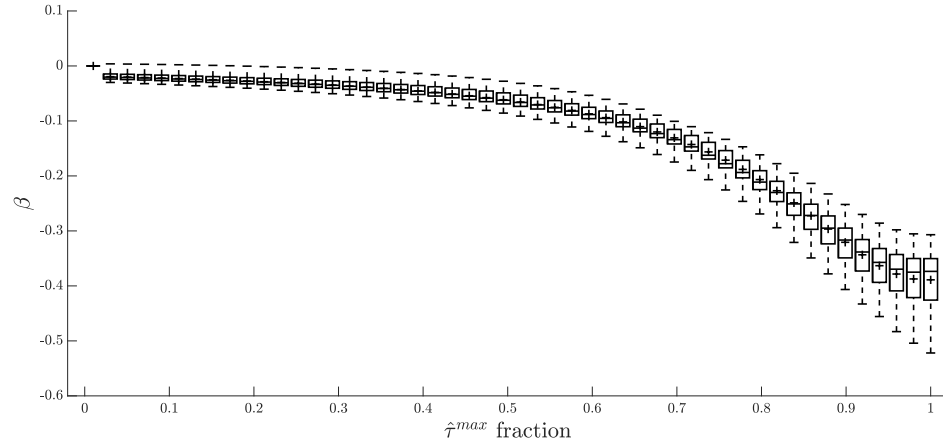
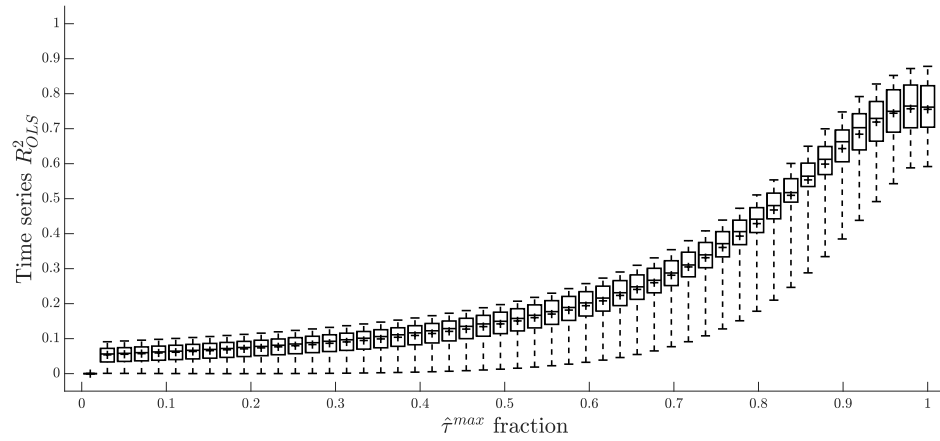


Figure OA-2: **Tradeoff between time series and cross-sectional S-SDF explanatory power of APT S-SDFs, non APT S-SDFs and principal components based S-SDFs.** For various minimum variance S-SDFs, the figure reports the tradeoff between average time series R^2 metric (32) and cross-sectional GLS R^2 metric (31) (denoted by R^2_{GLS} , left panels) in the main text, or cross-sectional R^2 metric (34) (denoted by R^2_{OLS} , right panels) in the main text. The following S-SDFs are compared in Panel (A). First, APT S-SDFs based on the l_2 -metric. Second, non APT S-SDFs based on the l_2 -metric, but with a diagonal weighting matrix $\Sigma_\eta = \mathbf{I}_{N_D \times N_D}$. Both these S-SDFs take the first principal component of excess returns as the single traded factor. In each figure of Panel (B), results for two additional principal components based S-SDFs are reported, which impose the corresponding pricing constraints on the principal components of sorted portfolio excess returns, instead of the original excess returns. The first and second of these S-SDFs impose in the left (right) figure an APT l_2 - and l_∞ - pricing constraint (a non APT l_2 - and l_∞ - pricing constraint based on a diagonal weighting matrix $\Sigma_\eta = \mathbf{I}_{N_D \times N_D}$), respectively. All results are based on the low dimensional dataset running from 1963 to 2018.



(1) Panel (A)



(2) Panel (B)

Figure OA-3: **Cross-sectional distributions of APT S–SDF betas and time series R^2 metrics.** The figure reports in **Panel (A)** (**Panel (B)**) the cross-sectional distribution of S–SDF betas (time series R^2 metrics (32) in the main text) as a function of threshold τ , for minimum variance S–SDFs satisfying APT pricing constraint (16) of the main text under an l_2 –metric, and while taking the market excess return as the single traded factor. In both panels, plus signs (horizontal lines) in the boxes indicate cross-sectional averages (medians). All computations are based on the low dimensional dataset from 1963 to 2018.

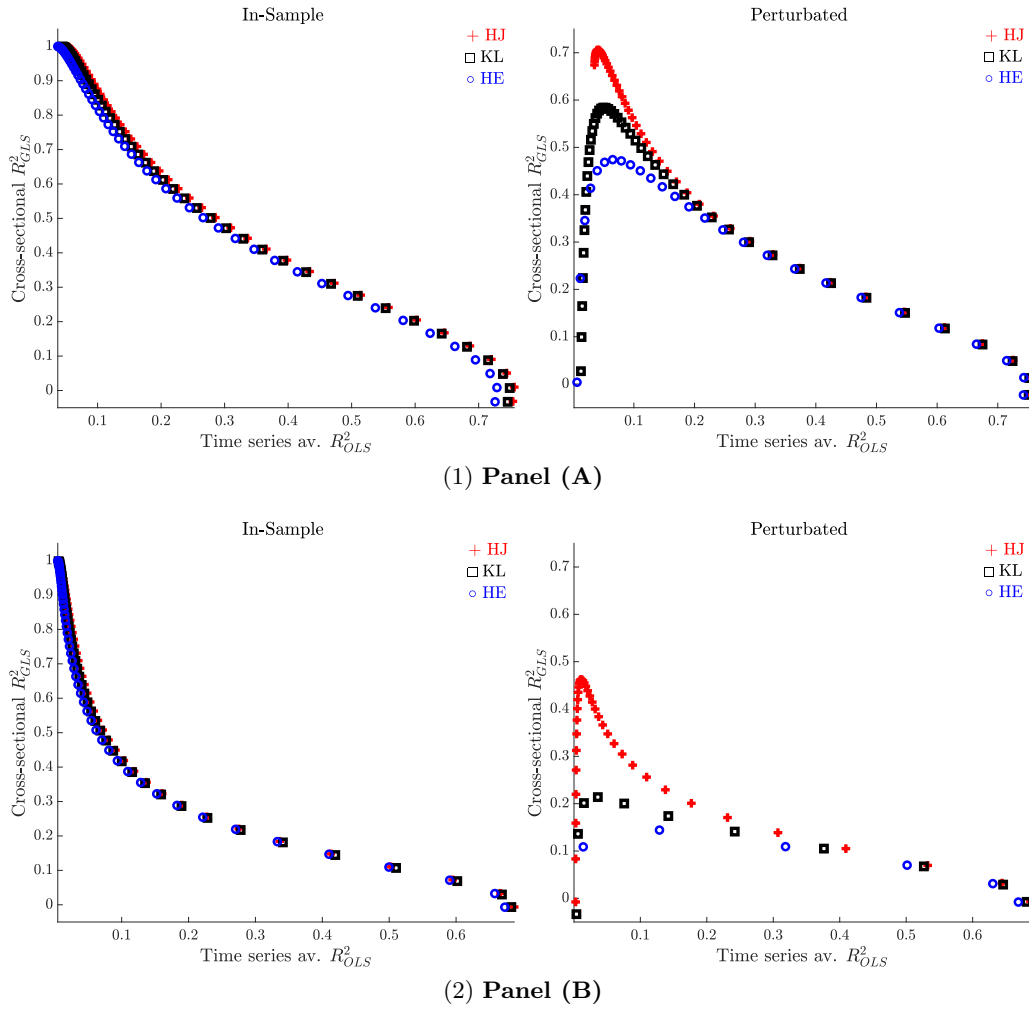


Figure OA-4: **Tradeoff between time series and cross-sectional APT S-SDF explanatory power, under different notions of S-SDF dispersion and data perturbations.** The figure reports the tradeoff between average time series R^2 metric (32) and cross-sectional GLS R^2 metric (31) in the main text for minimum dispersion APT S-SDFs based on the l_2 -metric and on different notions of S-SDF dispersion from the Cressie and Read [1984] family in Section B.1 of this Online Appendix: Variance (label HJ, case 3(a)), Kullback-Leibler divergence (label KL, case 1) and Hellinger divergence (label HE, case 3(b)). **Panel (A)** reports results for the low dimensional dataset and **Panel (B)** for the intermediate dimensional dataset. Left panels report in-sample results, where the estimation and evaluation periods overlap on the 40 (50) year window 1965-2005 (1955-2005) for the low (intermediate) dimensional dataset. For the low (intermediate) dimensional dataset, right panels report results for a 40 (50) year estimation window 1965-2005 (1955-2015) and a translated 40 (50) year evaluation window 1975-2015 (1965-2015) of 10 years.

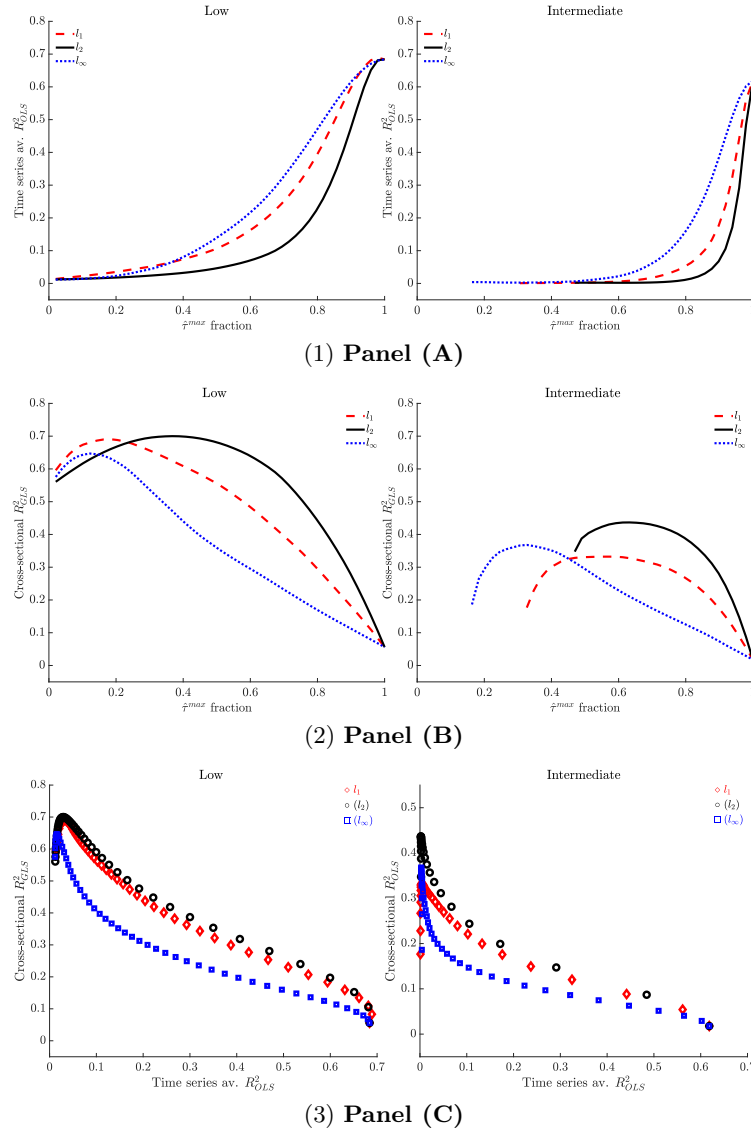


Figure OA-5: **Out-of-sample tradeoff between time-series and cross-sectional APT S-SDF explanatory power.** The figure reports the out-of-sample average time-series R^2 metric (32) (top panels) and the out-of-sample cross-sectional GLS R^2 metric (31) (middle panels) from the main text, for the sequence of minimum variance APT S-SDFs from Section C.2. These S-SDFs are estimated on rolling windows of 30 years under a constant relative threshold $\bar{\kappa} \in [\max_y \{\hat{\kappa}_y^{min}\}, 1]$ and based on a l_1 -, l_2 -, and (scaled) l_∞ -metric, respectively. In each estimation window, S-SDFs correctly price the risk-free asset and the market return. Bottom panels directly report the tradeoff between out-of-sample average time series R^2 metric (32) and out-of-sample cross-sectional GLS R^2 metric (31). Results are based on the low dimensional (left panels) and the intermediate dimensional (right panels) datasets.

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